

Statistics for Data Science II

Comprehensive Course Textbook & Problem Solver

IITM BS Degree Program in Data Science and Applications

August 4, 2026

Contents

Introduction	ii
1 Joint & Marginal Discrete PMFs, Conditional Distributions	1
1.1 Joint Probability Mass Function (Joint PMF)	1
1.2 Marginal PMFs	2
1.3 Conditional PMFs	2
1.4 Practice Exercises	3
2 Independence of RVs, Functions of RVs & PMF Convolution	4
2.1 Independence of Discrete Random Variables	4
2.2 Functions of Random Variables and PMF Convolution	5
2.3 Minimum and Maximum of Random Variables	5
2.4 Practice Exercises	6
3 Expectation, Covariance, Correlation & Probability Bounds	7
3.1 Expected Value and Linearity of Expectation	7
3.2 Covariance and Correlation	7
3.3 Markov and Chebyshev Inequalities	8
3.4 Practice Exercises	9
4 Continuous Random Variables, CDF, PDF & Standard Distributions	10
4.1 Cumulative Distribution Function (CDF) and Probability Density Function (PDF) .	10
4.2 Standard Continuous Distributions	11
4.3 Practice Exercises	12
5 Functions of Continuous RVs, Expectations & Mixed Distributions	13
5.1 Transformations of Continuous Random Variables	13
5.2 Expectation and Variance of Continuous RVs	14
5.3 Joint Distributions of Discrete and Continuous RVs	14
5.4 Practice Exercises	15
6 Joint Continuous RVs, Marginal Densities & Conditional Densities	16
6.1 Joint Probability Density Function	16
6.2 Marginal Densities and Independence	16
6.3 Conditional Densities and Conditional Expectation	17
6.4 Practice Exercises	18
7 Statistics from Samples & The Law of Large Numbers	19
7.1 i.i.d. Samples and Sample Statistics	19
7.2 Empirical Distribution Function	19
7.3 The Weak Law of Large Numbers (WLLN)	20
7.4 Practice Exercises	21

8	Moment Generating Functions & The Central Limit Theorem	22
8.1	Moment Generating Functions (MGF)	22
8.2	The Central Limit Theorem (CLT)	23
8.3	Practice Exercises	24
9	Parameter Estimation: MME, MLE & Confidence Intervals	25
9.1	Point Estimators: Bias, Variance, and MSE	25
9.2	Method of Moments Estimator (MME)	25
9.3	Maximum Likelihood Estimator (MLE)	26
9.4	Confidence Intervals	26
9.5	Practice Exercises	28
10	Bayesian Estimation: Prior, Posterior, MAP & Bayes Risk	29
10.1	The Bayesian Framework and Bayes' Rule for Parameters	29
10.2	Point Estimators: MAP and Posterior Mean	29
10.3	Practice Exercises	31
11	Hypothesis Testing: Size, Power, Z-Test & p-Values	32
11.1	Null and Alternative Hypotheses, Errors, and Power	32
11.2	The One-Sample Z-Test	32
11.3	The p -Value	33
11.4	Practice Exercises	34
12	Advanced Hypothesis Testing, LRT & Goodness-of-Fit	35
12.1	Student's t -Test for Variance Unknown	35
12.2	Two-Sample Tests	35
12.3	Likelihood Ratio Test (LRT) and Chi-Squared Goodness of Fit	36
12.4	Practice Exercises	38
	Solutions to Practice Exercises	40

Introduction

Welcome to **Statistics for Data Science 2 (SDS2)**. Building upon foundational descriptive statistics and univariate probability, this course delivers advanced statistical inference, joint probability distributions, parameter estimation (MLE, MME, Bayesian MAP), Central Limit Theorem applications, and hypothesis testing frameworks.

Structure of this Guide

The course is divided into **12 Chapters**:

- **Chapters 1–4 (Quiz 1 Content):** Joint & Marginal Discrete/Continuous Distributions, Independence, Functions of RVs, Covariance, and Correlation.
- **Chapters 5–8 (Quiz 2 Content):** Expectations of Functions, Mixed Distributions, Marginal/-Conditional Densities, Sample Statistics, Law of Large Numbers (LLN), Moment Generating Functions (MGF), and Central Limit Theorem (CLT).
- **Chapters 9–12 (End-Term Content):** Method of Moments (MME), Maximum Likelihood Estimation (MLE), Bayesian Inference (Prior/Posterior/MAP), Confidence Intervals, Z/t Hypothesis Testing, and Likelihood Ratio Tests (LRT).

Visual Layout & Box Legend

- **Emerald Cards** — Definitions and statistical concepts.
- **Teal Cards** — Statistical theorems, LLN/CLT properties, and rules.
- **Slate Cards** — Worked numerical estimation & hypothesis testing examples.
- **Mint/Teal Cards** — Core intuition and probabilistic reasoning.
- **Green Cards** — Key statistical formulas and estimators.
- **Emerald Video Cards** — NPTEL video lecture resource links.

Chapter 1

Joint & Marginal Discrete PMFs, Conditional Distributions

In real-world data science, outcomes rarely depend on a single variable in isolation. We often observe pairs or vectors of random variables simultaneously. In this chapter, we extend single-variable probability concepts to **Joint Probability Mass Functions (Joint PMFs)**, **Marginal Distributions**, and **Conditional Distributions** for discrete random variables.

1.1 Joint Probability Mass Function (Joint PMF)

When two discrete random variables X and Y are defined on the same sample space, their combined behavior is specified by their **Joint PMF**.

Definition: Joint PMF $p_{X,Y}(x, y)$

The **Joint Probability Mass Function** of discrete random variables X and Y is defined as:

$$p_{X,Y}(x, y) = \mathbb{P}(X = x \text{ and } Y = y) = \mathbb{P}(X = x, Y = y)$$

A valid joint PMF must satisfy two fundamental conditions:

1. Non-negativity: $p_{X,Y}(x, y) \geq 0$ for all (x, y) .
2. Normalization: $\sum_x \sum_y p_{X,Y}(x, y) = 1$.

Worked Example: Two Dice Roll Joint PMF

Let X be the number shown on die 1 and Y be the number on die 2. Since the dice are independent and fair:

$$p_{X,Y}(x, y) = \frac{1}{36} \quad \text{for } x \in \{1, \dots, 6\}, y \in \{1, \dots, 6\}$$

The sum of all 36 probabilities is $36 \times \frac{1}{36} = 1$.

Video Resource: IITM BS Lecture: Joint PMF of Two Discrete Random Variables



Official lecture introducing joint probability tables, bivariate discrete distributions, and properties.

1.2 Marginal PMFs

Given the joint PMF $p_{X,Y}(x, y)$, we can recover the individual distribution of X alone (or Y alone) by summing over all possible values of the other variable. This process is called **Marginalization**.

Key Formula: Marginal PMF Formulas

- **Marginal PMF of X :**

$$p_X(x) = \mathbb{P}(X = x) = \sum_y p_{X,Y}(x, y)$$

- **Marginal PMF of Y :**

$$p_Y(y) = \mathbb{P}(Y = y) = \sum_x p_{X,Y}(x, y)$$

Data Science Relevance: In tabular data (e.g. customer churn vs age bracket), computing column totals or row totals in a contingency table corresponds directly to finding marginal PMFs!

Video Resource: IITM BS Lecture: Marginal PMF of Discrete Random Variables



Official lecture deriving row/column marginal distributions from joint probability tables.

1.3 Conditional PMFs

How does knowing the value of Y update our belief about the distribution of X ? This is captured by the **Conditional PMF**.

Definition: Conditional PMF $p_{X|Y}(x | y)$

For any y such that $p_Y(y) > 0$, the **Conditional PMF** of X given $Y = y$ is:

$$p_{X|Y}(x | y) = \mathbb{P}(X = x | Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

For a fixed y , $\sum_x p_{X|Y}(x | y) = 1$, making $p_{X|Y}(\cdot | y)$ a valid probability distribution over x .

Video Resource: IITM BS Lecture: Conditional Distribution of One Random Variable Given Another

Official lecture defining conditional PMFs and constructing conditional probability slices.

Video Resource: IITM BS Lecture: Joint PMF of More Than Two Random Variables



Official lecture extending joint PMFs and conditional distributions to multivariate vectors $(X_1, X_2, \dots, X_n)$.

1.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Joint PMF Table Verification:** The joint PMF of $X \in \{0, 1\}$ and $Y \in \{1, 2\}$ is given by the table:

$p_{X,Y}(x, y)$	$Y = 1$	$Y = 2$
$X = 0$	0.1	0.3
$X = 1$	0.2	k

Find the value of constant k so that $p_{X,Y}$ is a valid joint PMF.

2. **Marginal Distribution Computation:** Using the completed table from Exercise 1:
 - (a) Find the marginal PMF $p_X(x)$ for $x \in \{0, 1\}$.
 - (b) Find the marginal PMF $p_Y(y)$ for $y \in \{1, 2\}$.
3. **Conditional Probability Calculation:** Calculate $p_{X|Y}(1 | 2)$, the conditional probability that $X = 1$ given $Y = 2$.
4. **Triangular Domain Joint PMF:** Let X and Y have joint PMF $p_{X,Y}(x, y) = c(x + y)$ for $x \in \{1, 2\}$ and $y \in \{1, 2\}$.
 - (a) Find the normalization constant c .
 - (b) Find $p_X(1)$.
5. **Data Science Application (A/B Testing Click-Through Distribution):** In an online ad campaign, $X \in \{0, 1\}$ represents device type (0 = Mobile, 1 = Desktop) and $Y \in \{0, 1\}$ represents click outcome (0 = No Click, 1 = Click). Given joint probabilities:

$$p_{X,Y}(0, 0) = 0.50, \quad p_{X,Y}(0, 1) = 0.10, \quad p_{X,Y}(1, 0) = 0.32, \quad p_{X,Y}(1, 1) = 0.08$$

Compare the Click-Through Rate (CTR) for Mobile users $\mathbb{P}(Y = 1 | X = 0)$ versus Desktop users $\mathbb{P}(Y = 1 | X = 1)$.

Chapter 2

Independence of RVs, Functions of RVs & PMF Convolution

Independence is the bedrock of statistical modeling and machine learning assumptions (e.g. Naive Bayes, i.i.d. sampling). In this chapter, we master **Independence of Random Variables, Functions of Random Variables, PMF Convolution** for sums of independent RVs, and distributions of **Extreme Values** ($\min(X, Y)$ and $\max(X, Y)$).

2.1 Independence of Discrete Random Variables

Two random variables X and Y are independent if knowing the value of Y gives zero information about the probability distribution of X .

Definition: Independence Criterion

Discrete random variables X and Y are **Independent** (written $X \perp\!\!\!\perp Y$) if and only if their joint PMF factorizes into the product of their marginal PMFs for all (x, y) :

$$p_{X,Y}(x, y) = p_X(x) \cdot p_Y(y) \quad \forall x, y$$

Equivalently, $p_{X|Y}(x | y) = p_X(x)$ for all y where $p_Y(y) > 0$.

Video Resource: IITM BS Lecture: Independence of Two Random Variables



Official lecture defining pairwise independence and testing factorization across joint PMF tables.

Video Resource: IITM BS Lecture: Independence of Multiple Random Variables



Official lecture extending independence to n random variables: $p_{X_1, \dots, X_n}(x_1, \dots, x_n) = \prod_{i=1}^n p_{X_i}(x_i)$.

2.2 Functions of Random Variables and PMF Convolution

If X and Y are random variables and $Z = g(X, Y)$, then Z is also a random variable. To find the PMF of Z , we sum the joint probabilities of all outcome pairs (x, y) that map to z :

$$p_Z(z) = \sum_{\{(x,y)|g(x,y)=z\}} p_{X,Y}(x, y)$$

Key Formula: Convolution Formula for Sum $Z = X + Y$

If X and Y are **independent** discrete random variables, the PMF of $Z = X + Y$ is the **Discrete Convolution** of p_X and p_Y :

$$p_Z(z) = \mathbb{P}(X + Y = z) = \sum_x p_X(x) \cdot p_Y(z - x)$$

Data Science Relevance: Sums of independent Poisson variables ($\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2) \sim \text{Poisson}(\lambda_1 + \lambda_2)$) and Binomial variables are used in modeling queuing arrivals and ensemble predictions!

Video Resource: IITM BS Lecture: Functions of Random Variables & PMF Convolution



Official lecture computing PMFs of transformations $g(X, Y)$ and deriving discrete convolutions.

2.3 Minimum and Maximum of Random Variables

Extreme values $V = \min(X, Y)$ and $W = \max(X, Y)$ appear frequently in system reliability, latency modeling, and auction theory.

Key Formula: Min and Max CDF Formulas for Independent RVs

For independent random variables X and Y :

- **Maximum** $W = \max(X, Y)$:

$$F_W(w) = \mathbb{P}(\max(X, Y) \leq w) = \mathbb{P}(X \leq w \text{ and } Y \leq w) = F_X(w) \cdot F_Y(w)$$

- **Minimum** $V = \min(X, Y)$:

$$\mathbb{P}(V > v) = \mathbb{P}(\min(X, Y) > v) = \mathbb{P}(X > v \text{ and } Y > v) = (1 - F_X(v)) \cdot (1 - F_Y(v))$$

Video Resource: IITM BS Lecture: Minimum & Maximum of Two Random Variables



Official lecture deriving probability mass functions and cumulative distributions for $\min(X, Y)$ and $\max(X, Y)$.

2.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Testing Independence from Joint PMF:** Let $X \in \{0, 1\}$ and $Y \in \{0, 1\}$ have joint PMF:

$$p_{X,Y}(0, 0) = 0.4, \quad p_{X,Y}(0, 1) = 0.2, \quad p_{X,Y}(1, 0) = 0.2, \quad p_{X,Y}(1, 1) = 0.2$$

Are X and Y independent? Justify by comparing $p_{X,Y}(0, 0)$ with $p_X(0) \cdot p_Y(0)$.

2. **PMF Convolution of Independent Uniforms:** Let X and Y be independent discrete uniform random variables on $\{1, 2, 3\}$. Find the PMF of $Z = X + Y$.
3. **Sum of Independent Poisson RVs:** Let $X \sim \text{Poisson}(\lambda_1 = 2)$ and $Y \sim \text{Poisson}(\lambda_2 = 3)$ be independent. State the distribution of $Z = X + Y$ and compute $\mathbb{P}(Z = 1)$.
4. **Maximum of Two Geometric RVs:** Let X and Y be independent geometric random variables with success probability $p = 0.5$ ($\mathbb{P}(X \leq k) = 1 - (1 - p)^k$). Find the CDF $F_W(2)$ for $W = \max(X, Y)$.
5. **Data Science Application (Server Latency Redundancy Minimum):** A high-availability web app routes queries to two parallel microservices. Let $X, Y \in \{1, 2, 3\}$ seconds be independent service response times with $p_X(k) = p_Y(k) = 1/3$. Compute the PMF of the overall system response time $V = \min(X, Y)$.

Chapter 3

Expectation, Covariance, Correlation & Probability Bounds

Summary statistics such as the mean (center) and variance (spread) compress complex probability distributions into single numbers. In this chapter, we master **Joint Expectation**, **Covariance**, Pearson's **Correlation Coefficient**, and distribution-free probability bounds via **Markov's** and **Chebyshev's Inequalities**.

3.1 Expected Value and Linearity of Expectation

Definition: Expected Value $\mathbb{E}[g(X, Y)]$

For joint discrete random variables X and Y , the expected value of any function $g(X, Y)$ is:

$$\mathbb{E}[g(X, Y)] = \sum_x \sum_y g(x, y) \cdot p_{X,Y}(x, y)$$

Theorem / Property: Linearity of Expectation

For any random variables $X_1, X_2, \dots, X_n$ (whether independent or dependent!) and constants a_i, b :

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i + b\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i] + b$$

If X and Y are **independent**, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Video Resource: IITM BS Lecture: Expected Value & Properties of Expectation



Official lecture introducing expectation for joint distributions and proving linearity of expectation.

3.2 Covariance and Correlation

To measure how two random variables co-vary together, we define **Covariance** and normalized **Correlation**.

Definition: Covariance $\text{Cov}(X, Y)$

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

- If X and Y are independent $\implies \text{Cov}(X, Y) = 0$. (Note: Zero covariance does **not** guarantee independence!)
- Variance of a sum: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$.

Key Formula: Pearson Correlation Coefficient $\rho_{X,Y}$

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad \text{where } \sigma_X = \sqrt{\text{Var}(X)}, \sigma_Y = \sqrt{\text{Var}(Y)}$$

- Always bounded: $-1 \leq \rho_{X,Y} \leq 1$.
- $\rho = +1$: Perfect positive linear relationship; $\rho = -1$: Perfect negative linear relationship.

Data Science Relevance: Feature correlation matrices are inspected before building linear regression models to detect multicollinearity, and covariance matrices Σ form the foundation of Principal Component Analysis (PCA)!

Video Resource: IITM BS Lecture: Covariance & Correlation Coefficient


Official lecture deriving covariance properties, variance of linear combinations, and correlation bounds.

3.3 Markov and Chebyshev Inequalities

When the exact probability distribution is unknown, we can bound tail probabilities using only the mean and variance.

Theorem / Property: Markov's Inequality

For a non-negative random variable $X \geq 0$ with finite mean $\mu = \mathbb{E}[X]$:

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a} \quad \text{for any } a > 0$$

Theorem / Property: Chebyshev's Inequality

For any random variable X with mean μ and variance σ^2 :

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2} \quad \text{for any } k > 0$$

Equivalently: $\mathbb{P}(|X - \mu| \geq \epsilon) \leq \frac{\text{Var}(X)}{\epsilon^2}$.

Video Resource: IITM BS Lecture: Bounds in Probabilities (Markov & Chebyshev)


Official lecture proving Markov's and Chebyshev's inequalities and applying them to tail probability bounds.

3.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Linearity of Expectation:** Roll 10 fair 6-sided dice. Let X_i be the score on die i , and $S = \sum_{i=1}^{10} X_i$ be the total sum. Find $\mathbb{E}[S]$.

2. **Covariance Calculation from Joint PMF:** The joint PMF of $X \in \{0, 1\}$ and $Y \in \{1, 2\}$ is:

$$p_{X,Y}(0, 1) = 0.2, \quad p_{X,Y}(0, 2) = 0.3, \quad p_{X,Y}(1, 1) = 0.1, \quad p_{X,Y}(1, 2) = 0.4$$

Compute $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathbb{E}[XY]$, and the covariance $\text{Cov}(X, Y)$.

3. **Correlation Coefficient:** For the variables X and Y in Exercise 2:

- (a) Compute $\text{Var}(X)$ and $\text{Var}(Y)$.
- (b) Compute the Pearson correlation coefficient $\rho_{X,Y}$.

4. **Markov's Inequality Tail Bound:** The average daily website visits for a blog is $\mathbb{E}[X] = 500$. Using Markov's inequality, find an upper bound for the probability that on a given day the blog receives at least 2,000 visits ($\mathbb{P}(X \geq 2000)$).
5. **Data Science Application (Algorithmic Trading Risk Bound):** An automated trading strategy has expected daily return $\mu = \$100$ with standard deviation $\sigma = \$20$. Use Chebyshev's inequality to find a lower bound for the probability that daily return falls between \$60 and \$140.

Chapter 4

Continuous Random Variables, CDF, PDF & Standard Distributions

Unlike discrete random variables taking distinct point values, **Continuous Random Variables** can take any real value in an interval. In this chapter, we master the **Cumulative Distribution Function (CDF)**, **Probability Density Function (PDF)**, and the fundamental benchmark distributions: **Uniform**, **Exponential**, and **Gaussian (Normal)**.

4.1 Cumulative Distribution Function (CDF) and Probability Density Function (PDF)

Definition: Cumulative Distribution Function $F_X(x)$

The **Cumulative Distribution Function (CDF)** of any random variable X is defined for all $x \in \mathbb{R}$ by:

$$F_X(x) = \mathbb{P}(X \leq x)$$

Properties of $F_X(x)$:

1. Non-decreasing: $x_1 < x_2 \implies F_X(x_1) \leq F_X(x_2)$.
2. Limits: $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$.
3. Right-continuous: $\lim_{h \rightarrow 0^+} F_X(x + h) = F_X(x)$.

Definition: Probability Density Function $f_X(x)$

A random variable X is **Continuous** if there exists a non-negative function $f_X(x) \geq 0$ such that for all x :

$$F_X(x) = \int_{-\infty}^x f_X(t) dt \implies f_X(x) = \frac{d}{dx} F_X(x)$$

Key Probability Rule: For continuous X , the probability at any single point is zero ($\mathbb{P}(X = c) = 0$). Interval probabilities are areas under the PDF curve:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx = F_X(b) - F_X(a)$$

Normalizing condition: $\int_{-\infty}^{\infty} f_X(x) dx = 1$.

Video Resource: IITM BS Lecture: Cumulative Distribution Function (CDF)

Official lecture defining CDF properties, right-continuity, and interval probability calculation.

Video Resource: IITM BS Lecture: Probability Density Function (PDF)

Official lecture introducing continuous densities $f_X(x)$ and integration rules.

4.2 Standard Continuous Distributions

Key Formula: 1. Uniform Distribution $\text{Uniform}(a, b)$

- **PDF:** $f(x) = \frac{1}{b-a}$ for $x \in [a, b]$.
- **CDF:** $F(x) = \frac{x-a}{b-a}$ for $x \in [a, b]$.
- **Mean & Variance:** $\mathbb{E}[X] = \frac{a+b}{2}$, $\text{Var}(X) = \frac{(b-a)^2}{12}$.

Key Formula: 2. Exponential Distribution $\text{Exponential}(\lambda)$

- **PDF:** $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ ($\lambda > 0$).
- **CDF:** $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$.
- **Mean & Variance:** $\mathbb{E}[X] = \frac{1}{\lambda}$, $\text{Var}(X) = \frac{1}{\lambda^2}$.
- **Memoryless Property:** $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$.

Key Formula: 3. Normal (Gaussian) Distribution $\mathcal{N}(\mu, \sigma^2)$

- **PDF:** $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ for $x \in \mathbb{R}$.
- **Standard Normal** $Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0, 1)$: CDF denoted $\Phi(z) = \mathbb{P}(Z \leq z)$.

Data Science Relevance: Normal distributions appear in Gaussian Naive Bayes, Linear Regression noise assumptions ($\epsilon \sim \mathcal{N}(0, \sigma^2)$), and standardized feature scaling (Z-score normalization $z = \frac{x-\mu}{\sigma}$)!

Video Resource: IITM BS Lecture: Common Continuous Distributions (Uniform, Exponential, Normal)

Official lecture deriving properties of uniform, exponential, and Gaussian random variables.

4.3 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **PDF Normalization and Probability:** A continuous random variable X has PDF:

$$f_X(x) = \begin{cases} cx^2 & \text{for } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the constant c .
 - (b) Compute $\mathbb{P}(1 \leq X \leq 2)$.
2. **CDF to PDF Conversion:** The CDF of a continuous random variable X is:

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - e^{-3x} & \text{if } x \geq 0 \end{cases}$$

Identify the distribution of X , write its PDF $f_X(x)$, and calculate $\mathbb{E}[X]$.

3. **Exponential Memoryless Property:** Let $T \sim \text{Exponential}(\lambda = 0.5)$ represent customer service call duration in minutes. Compute $\mathbb{P}(T > 4 \mid T > 2)$.
4. **Standard Normal Z-Score Calculation:** Let $X \sim \mathcal{N}(\mu = 50, \sigma^2 = 100)$. Given $\Phi(1.5) = 0.9332$, compute $\mathbb{P}(X \geq 65)$.
5. **Data Science Application (Feature Normalization Range):** A model feature X follows a $\text{Uniform}(10, 50)$ distribution.
- (a) Find the mean μ_X and standard deviation σ_X . 2. Compute the probability that a random feature observation falls within 1 standard deviation of the mean ($\mathbb{P}(\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X)$).

Chapter 5

Functions of Continuous RVs, Expectations & Mixed Distributions

When we apply a mathematical function $Y = g(X)$ to a continuous random variable X , its probability density changes. In this chapter, we master the **CDF/PDF Methods for Transformations of Continuous RVs**, **Continuous Expectation** ($\mathbb{E}[g(X)]$), and **Joint Discrete-Continuous Distributions**.

5.1 Transformations of Continuous Random Variables

To determine the PDF of $Y = g(X)$ given $f_X(x)$:

Key Formula: The CDF Method for Transformations

1. Compute the CDF of Y :

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y) = \int_{\{x|g(x) \leq y\}} f_X(x) dx$$

2. Differentiate $F_Y(y)$ with respect to y to get $f_Y(y) = \frac{d}{dy} F_Y(y)$.

Theorem / Property: Change of Variables Formula for Monotonic $g(X)$

If $g(x)$ is strictly monotonic and differentiable with inverse $x = g^{-1}(y)$:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|$$

Worked Example: Linear Transformation $Y = aX + b$

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $Y = aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.

Video Resource: IITM BS Lecture: Functions of Continuous Random Variables (CDF & PDF Methods)

Official lecture deriving PDFs of transformed continuous random variables using CDF differentiation.

5.2 Expectation and Variance of Continuous RVs

Definition: Expectation $\mathbb{E}[g(X)]$ (LOTUS - Law of the Unconscious Statistician)

For a continuous random variable X with PDF $f_X(x)$:

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Special cases:

- Mean: $\mu = \mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$.
- Variance: $\sigma^2 = \text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Video Resource: IITM BS Lecture: Expectations of Continuous Random Variables



Official lecture computing continuous means, variances, and applying LOTUS.

5.3 Joint Distributions of Discrete and Continuous RVs

In many practical machine learning problems (e.g. classification), we observe a discrete class label $Y \in \{0, 1\}$ alongside continuous feature vector $X \in \mathbb{R}$.

Definition: Joint Discrete-Continuous PMF/PDF

Let N be a discrete RV ($p_N(n)$) and X be a continuous RV ($f_X(x)$).

- **Conditional PDF of X given $N = n$:** $f_{X|N}(x | n) = \frac{f_{X,N}(x,n)}{p_N(n)}$.
- **Law of Total Probability:**

$$f_X(x) = \sum_n f_{X|N}(x | n) \cdot p_N(n)$$

Data Science Relevance: The Law of Total Probability for mixed distributions forms the mathematical backbone of **Gaussian Mixture Models (GMM)** and generative classification models (Linear Discriminant Analysis)!

Video Resource: IITM BS Lecture: Joint Distribution of Discrete and Continuous RVs



Official lecture modeling mixed discrete-continuous pairs and computing marginal densities.

5.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **CDF Transformation Method:** Let $X \sim \text{Uniform}(0, 1)$ with $f_X(x) = 1$ for $x \in (0, 1)$. Find the PDF $f_Y(y)$ of $Y = -\ln(X)$ for $y > 0$. Identify the distribution of Y .
2. **Linear Transformation of Normal RV:** Let $X \sim \mathcal{N}(0, 1)$ be a standard normal random variable. Define $Y = 3X + 5$. Write the PDF of Y and state its mean and variance.
3. **Continuous LOTUS Expectation:** Let X have PDF $f_X(x) = 2x$ for $0 \leq x \leq 1$. Compute $\mathbb{E}[X^2]$.
4. **Variance of Uniform RV:** Compute $\text{Var}(X)$ for $X \sim \text{Uniform}(0, 4)$ using the integral definition $\mathbb{E}[X^2] - (\mathbb{E}[X])^2$.
5. **Data Science Application (Gaussian Mixture Model Marginal PDF):** A binary classifier models customer transaction amount X conditional on fraud status $N \in \{0, 1\}$.
 - Prior fraud probability: $p_N(1) = 0.05$, $p_N(0) = 0.95$.
 - Non-fraud distribution: $X \mid N = 0 \sim \mathcal{N}(\mu_0 = 50, \sigma_0^2 = 100)$.
 - Fraud distribution: $X \mid N = 1 \sim \mathcal{N}(\mu_1 = 300, \sigma_1^2 = 400)$.

Write the overall marginal PDF expression $f_X(x)$ for transaction amounts across all customers.

Chapter 6

Joint Continuous RVs, Marginal Densities & Conditional Densities

Extending single continuous distributions to multiple continuous random variables requires 2D double integrals over regions in $\mathbb{R}^2$. In this chapter, we master **Joint PDFs** $f_{X,Y}(x,y)$, **Marginal Densities**, **Continuous Independence**, and **Conditional Densities** $f_{X|Y}(x|y)$.

6.1 Joint Probability Density Function

Definition: Joint PDF $f_{X,Y}(x,y)$

Two continuous random variables X and Y have a **Joint PDF** $f_{X,Y}(x,y) \geq 0$ such that for any region $R \subseteq \mathbb{R}^2$:

$$\mathbb{P}((X,Y) \in R) = \iint_R f_{X,Y}(x,y) dx dy$$

Normalization Condition:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx dy = 1$$

Video Resource: IITM BS Lecture: Joint Continuous Random Variables

Official lecture introducing 2D density surfaces, double integrals over regions, and normalization.

6.2 Marginal Densities and Independence

Key Formula: Marginal Density Formulas

To extract the individual PDF of X (or Y) from the joint PDF $f_{X,Y}(x,y)$, integrate out the other variable:

- **Marginal PDF of X :** $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$
- **Marginal PDF of Y :** $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$

Definition: Continuous Independence Criterion

Continuous random variables X and Y are **Independent** ($X \perp\!\!\!\perp Y$) if and only if:

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y) \quad \text{for all } (x, y)$$

Video Resource: IITM BS Lecture: Marginal Densities of Continuous Random Variables

Official lecture deriving marginal PDF slices via partial integration.

Video Resource: IITM BS Lecture: Independence of Multiple Continuous Random Variables

Official lecture testing product factorization for joint continuous density functions.

6.3 Conditional Densities and Conditional Expectation

Definition: Conditional PDF $f_{X|Y}(x | y)$

For any y where $f_Y(y) > 0$, the **Conditional Density** of X given $Y = y$ is:

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

Conditional Expectation: $\mathbb{E}[X | Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx.$

Data Science Relevance: Conditional expectation $\mathbb{E}[Y | X = \mathbf{x}]$ is the mathematically optimal prediction function under Mean Squared Error loss forming the foundational goal of Regression Trees, Kernel Regression, and Neural Networks!

Video Resource: IITM BS Lecture: Conditional Density of Continuous Random Variables

Official lecture constructing conditional density curves and computing conditional expectations.

6.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Joint Density Normalization Constant:** Let X and Y have joint PDF:

$$f_{X,Y}(x,y) = \begin{cases} cxy & \text{for } 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Find constant c and compute $\mathbb{P}(X \leq 0.5, Y \leq 1)$.

2. **Marginal Densities Computation:** For the joint PDF $f_{X,Y}(x,y) = cxy$ on $0 \leq x \leq 1, 0 \leq y \leq 2$ from Exercise 1:
 - (a) Compute marginal density $f_X(x)$.
 - (b) Compute marginal density $f_Y(y)$.
 - (c) Are X and Y independent?
3. **Triangular Region Joint PDF:** Let X and Y be uniformly distributed over the triangle $0 \leq y \leq x \leq 1$ with joint PDF $f_{X,Y}(x,y) = 2$ for $0 \leq y \leq x \leq 1$. Find marginal density $f_X(x)$ and $f_Y(y)$.
4. **Conditional Density Calculation:** For the triangular joint PDF in Exercise 3: Find conditional density $f_{Y|X}(y|x)$ and calculate conditional expectation $\mathbb{E}[Y|X=x]$.
5. **Data Science Application (Bivariate Normal Feature Regression):** Standardized features (X,Y) follow a Bivariate Normal distribution with $\mu_X = 0, \mu_Y = 0, \sigma_X = 1, \sigma_Y = 1$ and correlation $\rho = 0.8$. Given the formula $\mathbb{E}[Y|X=x] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X)$, predict the expected value of Y when feature $X = 2.5$.

Chapter 7

Statistics from Samples & The Law of Large Numbers

Data science begins when we draw a finite sample of observations $X_1, X_2, \dots, X_n$ from an unknown underlying population. In this chapter, we master **Independent and Identically Distributed (i.i.d.) Samples**, the **Sample Mean** $\bar{X}_n$, the **Empirical CDF**, and the **Weak Law of Large Numbers (WLLN)**.

7.1 i.i.d. Samples and Sample Statistics

Definition: i.i.d. Samples

Random variables $X_1, X_2, \dots, X_n$ are **Independent and Identically Distributed (i.i.d.)** if:

1. Each X_i has the exact same marginal probability distribution (same mean μ and variance σ^2).
2. The joint distribution factorizes: $f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \prod_{i=1}^n f(x_i)$.

Key Formula: Sample Mean $\bar{X}_n$ Properties

For an i.i.d. sample $X_1, \dots, X_n$ with mean μ and variance σ^2 :

- **Sample Mean:** $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.
- **Unbiased Expectation:** $\mathbb{E}[\bar{X}_n] = \mu$.
- **Variance Vanishing Rate:** $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n} \implies \text{Standard Error } \text{SE}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$.

Video Resource: IITM BS Lecture: Statistics from i.i.d. Samples



Official lecture defining i.i.d. random variables and calculating expectation/variance of sample sums.

7.2 Empirical Distribution Function

Given an observed sample $x_1, x_2, \dots, x_n$, we can estimate the true population CDF $F(x)$ without making parametric assumptions.

Definition: Empirical Cumulative Distribution Function (ECDF)

The **Empirical CDF** $\hat{F}_n(x)$ is the fraction of sample points less than or equal to x :

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(x_i \leq x)$$

where $\mathbb{I}(\cdot)$ is the indicator function. By the WLLN, $\hat{F}_n(x)$ converges pointwise to the true CDF $F(x)$ as $n \rightarrow \infty$.

Video Resource: IITM BS Lecture: Empirical Distribution and Sample Mean

Official lecture constructing ECDFs and analyzing non-parametric distribution estimation.

7.3 The Weak Law of Large Numbers (WLLN)

As the sample size n increases, the sample mean $\bar{X}_n$ concentrates tightly around the true population mean μ .

Theorem / Property: Weak Law of Large Numbers (WLLN)

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with finite mean μ and variance σ^2 . For any margin of error $\epsilon > 0$:

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) = 0 \iff \lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| < \epsilon) = 1$$

Proof via Chebyshev's Inequality:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \xrightarrow{n \rightarrow \infty} 0$$

Data Science Relevance: The Law of Large Numbers guarantees that Monte Carlo simulations, A/B test metric estimations, and empirical risk minimization in Machine Learning converge to the true underlying population values as dataset size n grows!

Video Resource: IITM BS Lecture: Concentration Phenomenon & Law of Large Numbers

Official lecture proving the Weak Law of Large Numbers using Chebyshev's inequality.

7.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Sample Mean Standard Error:** An i.i.d. sample of size $n = 100$ is drawn from a population with mean $\mu = 80$ and variance $\sigma^2 = 400$.
 - (a) Compute $\mathbb{E}[\bar{X}_{100}]$.
 - (b) Compute the standard error $\text{SE}(\bar{X}_{100}) = \sqrt{\text{Var}(\bar{X}_{100})}$.
2. **Chebyshev Sample Size Determination for WLLN:** How large a sample size n is required to ensure that the sample mean $\bar{X}_n$ lies within $\epsilon = 0.5$ units of the true mean μ with probability at least 95% ($\mathbb{P}(|\bar{X}_n - \mu| < 0.5) \geq 0.95$), if population variance $\sigma^2 = 4$?
3. **Empirical CDF Evaluation:** Given a sample of 5 observations: $x = [2.1, 4.5, 1.8, 6.0, 4.5]$.
 - (a) Calculate the value of the Empirical CDF $\hat{F}_5(4.0)$.
 - (b) Calculate $\hat{F}_5(4.5)$.
4. **Sum of i.i.d. Uniform RVs:** Let $X_1, X_2, \dots, X_{12}$ be i.i.d. $\text{Uniform}(0, 1)$ random variables. Let $S = \sum_{i=1}^{12} X_i$. Find $\mathbb{E}[S]$ and $\text{Var}(S)$.
5. **Data Science Application (Monte Carlo Pi Estimation Bounds):** In a Monte Carlo algorithm, n independent random points are dropped in a unit square. Let $I_i = 1$ if point i falls inside a unit circle ($\mathbb{P}(I_i = 1) = p = \pi/4 \approx 0.7854$) and 0 otherwise. Use Chebyshev's inequality to find the sample size n needed so that the Monte Carlo estimate $\hat{p}_n = \frac{1}{n} \sum_{i=1}^n I_i$ is within 0.01 of p with at least 90% probability.

Chapter 8

Moment Generating Functions & The Central Limit Theorem

Why does the Gaussian bell curve dominate statistics and machine learning? The answer lies in the **Central Limit Theorem (CLT)**. In this chapter, we master **Moment Generating Functions (MGF)** to track probability distributions and explore the asymptotic normality of sample averages.

8.1 Moment Generating Functions (MGF)

The **Moment Generating Function** $M_X(t)$ provides an elegant transform domain (analogous to Laplace/Fourier transforms) that uniquely characterizes a probability distribution.

Definition: Moment Generating Function $M_X(t)$

For a random variable X , its **Moment Generating Function (MGF)** is defined for t in an open interval around 0:

$$M_X(t) = \mathbb{E}[e^{tX}] = \begin{cases} \sum_x e^{tx} p_X(x) & \text{(Discrete)} \\ \int_{-\infty}^{\infty} e^{tx} f_X(x) dx & \text{(Continuous)} \end{cases}$$

Theorem / Property: Generating Moments and Distribution Uniqueness

1. **Moment Extraction:** Derivatives at $t = 0$ yield the raw moments $\mathbb{E}[X^k]$:

$$\left. \frac{d^k M_X(t)}{dt^k} \right|_{t=0} = \mathbb{E}[X^k] \implies M'_X(0) = \mathbb{E}[X], M''_X(0) = \mathbb{E}[X^2]$$

2. **Uniqueness Property:** If $M_X(t) = M_Y(t)$ for all t near 0, then X and Y have the exact same distribution! 3. **MGF of Independent Sums:** If $X \perp\!\!\!\perp Y$, then $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$.

Key Formula: Common MGF Formulas

- **Bernoulli(p):** $M(t) = (1 - p) + pe^t$.
- **Poisson(λ):** $M(t) = \exp(\lambda(e^t - 1))$.
- **Exponential(λ):** $M(t) = \frac{\lambda}{\lambda - t}$ for $t < \lambda$.

- **Normal**(μ, σ^2): $M(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$.

Video Resource: IITM BS Lecture: Moment Generating Functions (MGF)



Official lecture deriving MGFs, moment extraction derivatives, and MGF convolution for independent sums.

8.2 The Central Limit Theorem (CLT)

Theorem / Property: The Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with finite mean μ and variance $\sigma^2 > 0$. As $n \rightarrow \infty$, the normalized sample mean Z_n converges in distribution to the **Standard Normal Distribution** $\mathcal{N}(0, 1)$:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

For large n (typically $n \geq 30$), $\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ regardless of the shape of the original underlying distribution!

Data Science Relevance: CLT is the reason why z-confidence intervals, hypothesis tests, and normal error approximations in generalized linear models (GLMs) work reliably on large datasets even when features are non-Gaussian!

Video Resource: IITM BS Lecture: The Central Limit Theorem (CLT)



Official lecture proving CLT via MGF limit expansion $\lim_{n \rightarrow \infty} M_{Z_n}(t) = e^{t^2/2}$.

Video Resource: IITM BS Lecture: Descriptive Statistics of Normal Samples



Official lecture studying sample variance $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ and Chi-Squared distributions.

8.3 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **MGF Derivative Moment Extraction:** The MGF of a random variable X is $M_X(t) = \frac{1}{1-2t}$ for $t < 0.5$.
 - (a) Compute $\mathbb{E}[X] = M'_X(0)$.
 - (b) Compute $\mathbb{E}[X^2] = M''_X(0)$ and variance $\text{Var}(X)$.
2. **MGF of Independent Sums:** Let $X \sim \text{Poisson}(\lambda_1 = 3)$ and $Y \sim \text{Poisson}(\lambda_2 = 5)$ be independent. Use MGF product $M_{X+Y}(t) = M_X(t)M_Y(t)$ to prove that $X + Y \sim \text{Poisson}(8)$.
3. **CLT Probability Approximation for Sample Means:** A battery's lifetime in hours has mean $\mu = 40$ and standard deviation $\sigma = 8$. A company tests an i.i.d. sample of $n = 64$ batteries. Given $\Phi(1.5) = 0.9332$, use the CLT to approximate the probability that the sample mean lifetime $\bar{X}_{64}$ exceeds 41.5 hours.
4. **CLT Approximation for Binomial Counts:** A coin with probability of heads $p = 0.5$ is tossed $n = 100$ times. Use the Normal approximation (CLT) to find $\mathbb{P}(45 \leq S \leq 55)$ where $S = \sum_{i=1}^{100} X_i$. Given $\Phi(1.1) = 0.8643$.
5. **Data Science Application (A/B Test Conversion Sample Size):** An e-commerce platform compares conversion rates ($p = 0.10$, $\sigma = \sqrt{p(1-p)} = 0.30$). Using the CLT, how many visitor sessions n are required so that the standard error of the sample conversion rate $\text{SE}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$ is at most 0.005?

Chapter 9

Parameter Estimation: MME, MLE & Confidence Intervals

In Statistical Inference, we use observed data $x_1, \dots, x_n$ to estimate unknown population parameters θ . In this chapter, we master point estimation via **Method of Moments (MME)**, **Maximum Likelihood Estimation (MLE)**, evaluate estimator quality via **Bias and Variance**, and construct **Confidence Intervals**.

9.1 Point Estimators: Bias, Variance, and MSE

An **Estimator** $\hat{\theta}_n = g(X_1, \dots, X_n)$ is a statistic (a random variable) used to estimate unknown parameter θ .

Definition: Bias, Variance, and Mean Squared Error (MSE)

- **Bias:** $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$. An estimator is **Unbiased** if $\mathbb{E}[\hat{\theta}] = \theta$.
- **Variance:** $\text{Var}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2]$.
- **Mean Squared Error (MSE):**

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

Video Resource: IITM BS Lecture: Bias, Variance & Risk of an Estimator



Official lecture proving the Bias-Variance tradeoff decomposition for point estimators.

9.2 Method of Moments Estimator (MME)

The **Method of Moments** equates sample moments $\mu_k = \frac{1}{n} \sum_{i=1}^n X_i^k$ to population theoretical moments $\mathbb{E}[X^k] = g_k(\theta)$ and solves for θ .

Key Formula: Method of Moments Recipe

1. First Moment: Set $\bar{X} = \mathbb{E}[X]$. Solve for $\hat{\theta}_{\text{MME}}$.
2. Second Moment (if 2 parameters): Set $\frac{1}{n} \sum X_i^2 = \mathbb{E}[X^2]$.

Video Resource: IITM BS Lecture: Estimator Design: Method of Moments (MME)

Official lecture deriving MME point estimators for Poisson, Exponential, and Uniform parameters.

9.3 Maximum Likelihood Estimator (MLE)

The **Maximum Likelihood Estimator** finds the parameter value $\hat{\theta}_{\text{MLE}}$ that maximizes the probability density of observing the sample data.

Definition: Likelihood Function $L(\theta)$ and Log-Likelihood $\ell(\theta)$

For an i.i.d. sample $x_1, \dots, x_n$:

$$L(\theta) = \prod_{i=1}^n f(x_i | \theta) \implies \ell(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln f(x_i | \theta)$$

Solve the first-order optimization condition $\frac{d\ell(\theta)}{d\theta} = 0$ for $\hat{\theta}_{\text{MLE}}$.

Worked Example: Step-by-Step Derivation of Poisson MLE

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\lambda)$.

1. **Likelihood Function:** $L(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod x_i!}$.
2. **Log-Likelihood:** $\ell(\lambda) = \ln L(\lambda) = -n\lambda + (\sum_{i=1}^n x_i) \ln \lambda - \ln(\prod x_i!)$.
3. **First Derivative (Score Function):** $\frac{d\ell}{d\lambda} = -n + \frac{\sum x_i}{\lambda} = 0$.
4. **Solve for λ :** $\frac{\sum x_i}{\lambda} = n \implies \hat{\lambda}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{X}_n$.

Video Resource: IITM BS Lecture: Maximum Likelihood Estimation (MLE)

Official lecture constructing log-likelihood functions and solving likelihood equations.

9.4 Confidence Intervals

A $(1 - \alpha)$ **Confidence Interval** provides a range $[L, U]$ constructed from sample data that traps the true parameter θ with probability $1 - \alpha$.

Key Formula: Confidence Interval Formulas for Population Mean μ

- **Known Variance σ^2 (or large n via CLT):**

$$\left[\bar{X}_n - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

For 95% confidence ($\alpha = 0.05$), $z_{0.025} = 1.96$.

Data Science Relevance: Maximum Likelihood Estimation is the foundational optimization objective behind Logistic Regression, Gaussian Mixture Models, and Neural Network Cross-Entropy

loss minimization!

Video Resource: IITM BS Lecture: Confidence Intervals



Official lecture constructing two-sided z -confidence intervals and margin of error bounds.

9.5 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Unbiasedness Verification:** Let X_1, X_2, X_3 be an i.i.d. sample from a population with mean μ . Show that $\hat{\mu} = \frac{1}{4}X_1 + \frac{1}{2}X_2 + \frac{1}{4}X_3$ is an unbiased estimator for μ . Calculate its variance in terms of σ^2 .
2. **MME Parameter Estimation:** An i.i.d. sample of sizes $X_1, \dots, X_n$ is drawn from an Exponential(λ) distribution ($\mathbb{E}[X] = \frac{1}{\lambda}$). Derive the MME estimator $\hat{\lambda}_{\text{MME}}$.
3. **MLE Parameter Estimation:** An i.i.d. sample $x_1, \dots, x_n$ is drawn from a Poisson(λ) distribution ($f(x) = \frac{e^{-\lambda}\lambda^x}{x!}$). Derive the log-likelihood $\ell(\lambda)$ and prove that $\hat{\lambda}_{\text{MLE}} = \bar{X}_n$.
4. **95% Confidence Interval Construction:** A sample of $n = 100$ customer purchases has sample mean $\bar{X} = \$250$ with known population standard deviation $\sigma = \$40$. Construct a 95% confidence interval for the true mean purchase amount μ . (Use $z_{0.025} = 1.96$).
5. **Data Science Application (Logistic Regression MLE Loss):** In binary classification, $Y_i \in \{0, 1\}$ follows a Bernoulli(p_i) distribution where $p_i = \sigma(\mathbf{w}^T \mathbf{x}_i)$. Write out the negative log-likelihood loss function $J(\mathbf{w}) = -\ell(\mathbf{w})$ for an N -sample dataset (Binary Cross-Entropy Loss).

Chapter 10

Bayesian Estimation: Prior, Posterior, MAP & Bayes Risk

While Frequentist statistics treats parameters θ as fixed unknown constants, **Bayesian Statistics** treats θ as a random variable with its own probability distribution. In this chapter, we master **Prior Distributions, Bayes' Rule for Parameters, Posterior Distributions, Maximum A Posteriori (MAP) estimation, and Bayes Expected Loss/Risk.**

10.1 The Bayesian Framework and Bayes' Rule for Parameters

Definition: Prior, Likelihood, and Posterior

- **Prior Distribution** $f(\theta)$: Represents prior belief/knowledge about parameter θ before observing data.
- **Likelihood** $f(\mathbf{x} | \theta)$: The probability density of data $\mathbf{x} = (x_1, \dots, x_n)$ given parameter θ .
- **Posterior Distribution** $f(\theta | \mathbf{x})$: Updated probability density of θ after observing data $\mathbf{x}$:

$$f(\theta | \mathbf{x}) = \frac{f(\mathbf{x} | \theta)f(\theta)}{f(\mathbf{x})} = \frac{f(\mathbf{x} | \theta)f(\theta)}{\int f(\mathbf{x} | \theta')f(\theta') d\theta'} \propto L(\theta) \cdot f(\theta)$$

Video Resource: IITM BS Lecture: Bayesian Estimation



Official lecture introducing Bayesian philosophy, updating priors to posteriors using Bayes' rule.

Video Resource: IITM BS Lecture: Choice of Prior and Examples



Official lecture discussing uninformative priors, Beta-Binomial conjugate priors, and Normal-Normal conjugate priors.

10.2 Point Estimators: MAP and Posterior Mean

Once the posterior distribution $f(\theta | \mathbf{x})$ is computed, we extract point estimates for θ :

Key Formula: Bayesian Point Estimator Recipes

1. **Maximum A Posteriori (MAP) Estimator $\hat{\theta}_{\text{MAP}}$:** The mode of the posterior distribution (maximizes $f(\theta | \mathbf{x})$):

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} f(\theta | \mathbf{x}) = \arg \max_{\theta} [\ln f(\mathbf{x} | \theta) + \ln f(\theta)]$$

2. **Bayes Estimator (Posterior Mean $\hat{\theta}_{\text{Bayes}}$ under Squared Error Loss):** The expectation of the posterior distribution:

$$\hat{\theta}_{\text{Bayes}} = \mathbb{E}[\theta | \mathbf{x}] = \int \theta f(\theta | \mathbf{x}) d\theta$$

Data Science Relevance: MAP estimation is mathematically equivalent to **Regularized Machine Learning**! Specifically, MAP with a Gaussian prior is L_2 Ridge Regularization, while MAP with a Laplace prior is L_1 Lasso Regularization!

Video Resource: IITM BS Lecture: Bayesian Estimation: Finding Estimators



Official lecture solving MAP and Bayes Mean estimators for Beta-Binomial and Normal conjugate models.

10.3 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Beta-Binomial Conjugate Posterior:** Let coin flip success probability $p \sim \text{Beta}(\alpha = 2, \beta = 2)$ with prior PDF $f(p) = 6p(1 - p)$ for $0 \leq p \leq 1$. In $n = 5$ coin flips, $k = 4$ heads are observed ($L(p) = \binom{5}{4}p^4(1 - p)^1$).
 - (a) Find the posterior distribution $f(p \mid k = 4)$.
 - (b) Compute the MAP estimator $\hat{p}_{\text{MAP}}$.
 - (c) Compute the Bayes posterior mean estimator $\hat{p}_{\text{Bayes}}$.
2. **Normal-Normal Conjugate Model:** Let observation $X \mid \mu \sim \mathcal{N}(\mu, \sigma^2 = 4)$. A prior is placed on $\mu \sim \mathcal{N}(\mu_0 = 0, \sigma_0^2 = 1)$. Single observation $x = 5$ is made. Compute the posterior mean $\mathbb{E}[\mu \mid X = 5]$.
3. **Uniform Prior MAP vs MLE Comparison:** Let $X \sim \text{Binomial}(n = 10, p)$. If we assume a uniform prior $p \sim \text{Uniform}(0, 1)$, show that $\hat{p}_{\text{MAP}} = \hat{p}_{\text{MLE}} = \frac{x}{10}$.
4. **Laplace Prior and L_1 Regularization Equivalence:** Show that maximizing the log-posterior $\ln f(\mathbf{w} \mid \mathbf{x}, y) = \ln f(y \mid \mathbf{x}, \mathbf{w}) + \ln f(\mathbf{w})$ under a zero-mean Laplace prior $f(w_j) = \frac{\lambda}{2}e^{-\lambda|w_j|}$ adds an L_1 penalty term $-\lambda\|\mathbf{w}\|_1$ to the log-likelihood (Lasso regression).
5. **Data Science Application (Bayesian AB Testing Click Probability):** In an email campaign, variant A conversion probability p is assigned prior $\text{Beta}(1, 1)$ (Uniform). After sending $n = 100$ emails, $k = 15$ conversions occur. Compute the posterior distribution $\text{Beta}(\alpha', \beta')$ and the 95% posterior mean estimate for conversion rate.

Chapter 11

Hypothesis Testing: Size, Power, Z-Test & p-Values

Hypothesis testing provides a formal mathematical framework for decision-making under uncertainty. In this chapter, we master **Null vs. Alternative Hypotheses**, **Type I Error (α , Size)**, **Type II Error (β , Power $1 - \beta$)**, the **Z-Test**, and **p-values**.

11.1 Null and Alternative Hypotheses, Errors, and Power

Definition: Hypotheses and Decision Matrix

- **Null Hypothesis (H_0):** Baseline assumption of no effect or no difference (e.g. $H_0 : \mu = \mu_0$).
- **Alternative Hypothesis (H_1 or H_a):** Claim of interest we seek to prove (e.g. $H_1 : \mu \neq \mu_0$ or $H_1 : \mu > \mu_0$).

	Accept H_0 (Fail to Reject)	Reject H_0
H_0 True	Correct Decision ($1 - \alpha$)	Type I Error (α)
H_0 False	Type II Error (β)	Correct Decision (Power $1 - \beta$)

Key Formula: Size and Power of a Test

- **Significance Level / Size (α):** $\alpha = \mathbb{P}(\text{Reject } H_0 \mid H_0 \text{ is true})$.
- **Type II Error Probability (β):** $\beta = \mathbb{P}(\text{Fail to Reject } H_0 \mid H_0 \text{ is false})$.
- **Statistical Power ($1 - \beta$):** $1 - \beta = \mathbb{P}(\text{Reject } H_0 \mid H_1 \text{ is true})$.

Video Resource: IITM BS Lecture: Size and Power of a Test



Official lecture defining rejection regions, Type I/II errors, and calculating test power.

11.2 The One-Sample Z-Test

When sample size n is large (or population variance σ^2 is known), the sample mean $\bar{X}_n$ follows a Gaussian distribution under H_0 .

Key Formula: Z-Test Statistic and Rejection Regions

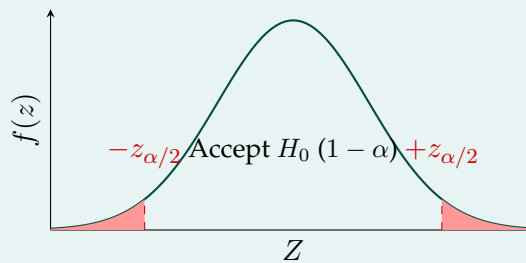
For $H_0 : \mu = \mu_0$ with test statistic $Z_{\text{calc}} = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}$:

- **Two-Tailed** ($H_1 : \mu \neq \mu_0$): Reject H_0 if $|Z_{\text{calc}}| \geq z_{\alpha/2}$.
- **Right-Tailed** ($H_1 : \mu > \mu_0$): Reject H_0 if $Z_{\text{calc}} \geq z_{\alpha}$.
- **Left-Tailed** ($H_1 : \mu < \mu_0$): Reject H_0 if $Z_{\text{calc}} \leq -z_{\alpha}$.

Common Critical Values: For $\alpha = 0.05 \implies z_{0.025} = 1.96$ (two-tailed), $z_{0.05} = 1.645$ (one-tailed).

Core Intuition: Visualizing Rejection Region and p -Value Tail Area

Below is the standard normal sampling distribution under H_0 , highlighting the critical rejection regions ($\alpha/2$) for a two-tailed test.

**Video Resource: IITM BS Lecture: Standard Testing Methods: Z-Test**

Official lecture executing one-sample and two-sample Z-tests on numerical dataset examples.

11.3 The p -Value

Definition: The p -Value Definition

The **p -value** is the probability of observing a test statistic at least as extreme as the one actually calculated from the sample, assuming H_0 is true.

Decision Rule: If $p\text{-value} \leq \alpha \implies \text{Reject } H_0$. If $p\text{-value} > \alpha \implies \text{Fail to Reject } H_0$.

Data Science Relevance: In A/B testing and feature selection (e.g. statsmodels OLS summary output), p -values tell us whether a feature's coefficient is statistically significantly different from zero ($\beta_j \neq 0$)!

Video Resource: IITM BS Lecture: The p -Value

Official lecture deriving p -values for right-tailed, left-tailed, and two-tailed tests.

11.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Type I and Type II Error Identification:** A medical diagnostic AI system evaluates H_0 : Patient is Healthy vs. H_1 : Patient has Disease.
 - (a) Explain what a Type I error represents in this context.
 - (b) Explain what a Type II error represents in this context. Which error is more dangerous?
2. **One-Sample Right-Tailed Z-Test:** An online store claims average order delivery time is at most $\mu_0 = 48$ hours. A customer protection group samples $n = 64$ deliveries and finds $\bar{X} = 51$ hours with known population standard deviation $\sigma = 12$ hours.
 - (a) State H_0 and H_1 .
 - (b) Compute the test statistic Z_{calc} .
 - (c) At significance level $\alpha = 0.05$ ($z_{0.05} = 1.645$), do we reject H_0 ?
3. **p-Value Calculation:** For the Z-test in Exercise 2 where $Z_{\text{calc}} = 2.0$: Given $\Phi(2.0) = 0.9772$, calculate the exact one-tailed p -value.
4. **Two-Tailed Z-Test with p -Value:** A machine fills cereal boxes with target weight $\mu_0 = 500\text{g}$. An inspector samples $n = 100$ boxes and finds $\bar{X} = 496\text{g}$ with $\sigma = 20\text{g}$. Compute Z_{calc} for $H_0 : \mu = 500$ vs $H_1 : \mu \neq 500$, calculate the two-tailed p -value (given $\Phi(2.0) = 0.9772$), and state the decision at $\alpha = 0.05$.
5. **Data Science Application (A/B Test Significance Decision):** An e-commerce A/B test compares new checkout UI (Treatment) vs current UI (Control). The test yields $Z_{\text{calc}} = 2.33$ for a two-tailed test. Given $\Phi(2.33) = 0.9901$, calculate the two-tailed p -value. Is the new checkout UI's conversion uplift statistically significant at $\alpha = 0.01$?

Chapter 12

Advanced Hypothesis Testing, LRT & Goodness of-Fit

In practical data science applications, population variance σ^2 is almost never known, requiring **Student's t -test**. Additionally, we frequently compare two populations or test if categorical data fits a theoretical model. In this final capstone chapter, we master **One-Sample and Two-Sample t -Tests**, **Chi-Squared (χ^2) Goodness-of-Fit Tests**, and the general **Likelihood Ratio Test (LRT)**.

12.1 Student's t -Test for Variance Unknown

When σ^2 is unknown, we substitute the sample variance $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.

Definition: One-Sample t -Test Statistic

For $H_0 : \mu = \mu_0$ with unknown σ^2 :

$$T_{\text{calc}} = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \quad (\text{Student's } t\text{-distribution with } n - 1 \text{ degrees of freedom})$$

The t -distribution has heavier tails than the standard normal $\mathcal{N}(0, 1)$, accounting for additional uncertainty in estimating S .

Video Resource: IITM BS Lecture: t -Test for Mean (Variance Unknown)



Official lecture deriving Student's t -test statistic, degrees of freedom, and critical value tables.

12.2 Two-Sample Tests

To test if two population means are equal ($H_0 : \mu_1 - \mu_2 = 0$):

Key Formula: Two-Sample Pooled t -Test Formula

Assuming equal variances $\sigma_1^2 = \sigma_2^2$:

$$T_{\text{calc}} = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

where pooled variance $S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$.

Video Resource: IITM BS Lecture: Two Samples from Normal Distribution



Official lecture deriving two-sample z -tests, pooled t -tests, and F -tests for variance equality.

12.3 Likelihood Ratio Test (LRT) and Chi-Squared Goodness of Fit

Definition: Likelihood Ratio Test (LRT)

For testing $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta \setminus \Theta_0$:

$$\Lambda(\mathbf{x}) = \frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)}$$

By Wilks' Theorem, $-2 \ln \Lambda(\mathbf{x}) \xrightarrow{d} \chi_k^2$ where $k = \dim(\Theta) - \dim(\Theta_0)$.

Key Formula: Pearson's Chi-Squared (χ^2) Goodness-of-Fit Test

To test whether categorical frequency data $O_1, \dots, O_k$ matches expected counts $E_1, \dots, E_k$:

$$\chi_{\text{calc}}^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi_{k-1-m}^2$$

Reject H_0 if $\chi_{\text{calc}}^2 \geq \chi_{\alpha, k-1-m}^2$.

```

1 import scipy.stats as stats
2 import numpy as np
3
4 # 1. One-Sample t-Test (H0: mu = 100)
5 data = np.array([102, 105, 98, 108, 101, 104, 99, 106, 103, 100, 107, 102, 105,
6                 101, 104, 98])
7 t_stat, p_val = stats.ttest_1samp(data, popmean=100)
8 print(f"One-Sample t-test: t = {t_stat:.3f}, p-value = {p_val:.4f}")
9
10 # 2. Two-Sample Independent t-Test (Pooled)
11 group1 = [14, 16, 15, 13, 17, 15, 14, 16, 15, 15]
12 group2 = [11, 13, 12, 10, 14, 12, 13, 11, 12, 12]
13 t_ind, p_ind = stats.ttest_ind(group1, group2, equal_var=True)
14 print(f"Two-Sample t-test: t = {t_ind:.3f}, p-value = {p_ind:.4f}")
15
16 # 3. Chi-Squared Goodness-of-Fit Test (Die Fairness)
17 observed = np.array([15, 7, 8, 12, 10, 8])
18 expected = np.array([10, 10, 10, 10, 10, 10])
19 chi2_stat, p_chi2 = stats.chisquare(f_obs=observed, f_exp=expected)
20 print(f"Chi-Squared test: chi2 = {chi2_stat:.3f}, p-value = {p_chi2:.4f}")

```

Listing 12.1: SciPy/Python Implementation of t-Tests and Chi-Squared Goodness of Fit

Video Resource: IITM BS Lecture: Likelihood Ratio Tests

Official lecture deriving likelihood ratios and Wilks' asymptotic χ^2 theorem.

Video Resource: IITM BS Lecture: Goodness of Fit for Discrete Distributions

Official lecture computing observed vs expected cell counts and applying χ^2 test statistic.

12.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **One-Sample t -Test Calculation:** A sample of $n = 16$ measurements has sample mean $\bar{X} = 104$ and sample standard deviation $S = 8$. We test $H_0 : \mu = 100$ vs $H_1 : \mu \neq 100$.

- (a) State degrees of freedom df .
- (b) Compute the test statistic T_{calc} .
- (c) At $\alpha = 0.05$ with critical value $t_{0.025,15} = 2.131$, state your decision.

2. **Two-Sample Pooled t -Test:** Two independent samples yield:

- Group 1: $n_1 = 10$, $\bar{X}_1 = 15$, $S_1^2 = 4$.
- Group 2: $n_2 = 10$, $\bar{X}_2 = 12$, $S_2^2 = 6$.

Compute pooled variance S_p^2 and T_{calc} for $H_0 : \mu_1 - \mu_2 = 0$.

3. **Chi-Squared Goodness of Fit (Die Fairness Test):** A 6-sided die is rolled $N = 60$ times, observing counts:

Observed $O_i = [15, 7, 8, 12, 10, 8]$ for faces $1 \dots 6$

Under H_0 : Die is fair ($E_i = 10$ for all i). Compute χ_{calc}^2 . At $\alpha = 0.05$ with critical value $\chi_{0.05,5}^2 = 11.07$, is the die fair?

4. **Wilks' Likelihood Ratio Test Stat:** A likelihood ratio test yields likelihood ratio $\Lambda = 0.05$. Compute Wilks' test statistic $W = -2 \ln \Lambda$.
5. **Comprehensive Course Review Question (χ^2 Independence Test):** In a marketing survey, 200 users are classified by Subscription Tier (Free, Premium) and Churn Status (Retained, Churned):

	Retained	Churned	Total
Free Tier	80 ($E = 90$)	40 ($E = 30$)	120
Premium Tier	70 ($E = 60$)	10 ($E = 20$)	80
Total	150	50	200

Compute the χ_{calc}^2 test statistic using the given expected counts E_{ij} to test for independence between Subscription Tier and Churn.

Index

Chi-Squared Test, 36

Likelihood Ratio Test, 36

t-Test, 36

Solutions to Practice Exercises

This appendix contains fully worked, step-by-step solutions for all practice exercises across Weeks 1 to 12.

Week 1: Joint & Marginal Discrete PMFs, Conditional Distributions

Detailed Solutions

1. Joint PMF Table Verification For $p_{X,Y}$ to be a valid joint PMF, the sum of all cell probabilities must equal 1:

$$\sum_{x=0}^1 \sum_{y=1}^2 p_{X,Y}(x, y) = 0.1 + 0.3 + 0.2 + k = 1 \implies 0.6 + k = 1 \implies \mathbf{k = 0.4}$$

2. Marginal Distribution Computation Using $k = 0.4$:

- **(a) Marginal PMF of X :**

$$p_X(0) = p_{X,Y}(0, 1) + p_{X,Y}(0, 2) = 0.1 + 0.3 = \mathbf{0.4}$$

$$p_X(1) = p_{X,Y}(1, 1) + p_{X,Y}(1, 2) = 0.2 + 0.4 = \mathbf{0.6}$$

- **(b) Marginal PMF of Y :**

$$p_Y(1) = p_{X,Y}(0, 1) + p_{X,Y}(1, 1) = 0.1 + 0.2 = \mathbf{0.3}$$

$$p_Y(2) = p_{X,Y}(0, 2) + p_{X,Y}(1, 2) = 0.3 + 0.4 = \mathbf{0.7}$$

3. Conditional Probability Calculation

$$p_{X|Y}(1 | 2) = \frac{p_{X,Y}(1, 2)}{p_Y(2)} = \frac{0.4}{0.7} = \frac{4}{7} \approx \mathbf{0.5714}$$

4. Triangular Domain Joint PMF

- **(a) Normalization Constant c :**

$$\sum_{x=1}^2 \sum_{y=1}^2 c(x+y) = c(1+1) + c(1+2) + c(2+1) + c(2+2) = 2c + 3c + 3c + 4c = 12c = 1 \implies \mathbf{c = \frac{1}{12}}$$

- **(b) Marginal $p_X(1)$:**

$$p_X(1) = p_{X,Y}(1, 1) + p_{X,Y}(1, 2) = 2c + 3c = 5c = 5 \left(\frac{1}{12} \right) = \mathbf{\frac{5}{12}}$$

5. Data Science Application: A/B Testing Click-Through Distribution

- **Mobile CTR** $\mathbb{P}(Y = 1 \mid X = 0)$:

$$p_X(0) = p_{X,Y}(0, 0) + p_{X,Y}(0, 1) = 0.50 + 0.10 = 0.60$$

$$\mathbb{P}(Y = 1 \mid X = 0) = \frac{p_{X,Y}(0, 1)}{p_X(0)} = \frac{0.10}{0.60} = \frac{1}{6} \approx \mathbf{16.67\%}$$

- **Desktop CTR** $\mathbb{P}(Y = 1 \mid X = 1)$:

$$p_X(1) = p_{X,Y}(1, 0) + p_{X,Y}(1, 1) = 0.32 + 0.08 = 0.40$$

$$\mathbb{P}(Y = 1 \mid X = 1) = \frac{p_{X,Y}(1, 1)}{p_X(1)} = \frac{0.08}{0.40} = \mathbf{0.20 = 20.00\%}$$

Desktop users yield a higher click-through rate (20.00%) than mobile users (16.67%).

Week 2: Independence of RVs, Functions of RVs & PMF Convolution

Detailed Solutions

1. Testing Independence from Joint PMF Compute marginal probabilities:

- $p_X(0) = p_{X,Y}(0, 0) + p_{X,Y}(0, 1) = 0.4 + 0.2 = 0.6$.
- $p_Y(0) = p_{X,Y}(0, 0) + p_{X,Y}(1, 0) = 0.4 + 0.2 = 0.6$.

Now check the product: $p_X(0) \cdot p_Y(0) = 0.6 \times 0.6 = 0.36$. Since $p_{X,Y}(0, 0) = 0.4 \neq 0.36$, **X and Y are NOT independent.**

2. PMF Convolution of Independent Uniforms $X, Y \in \{1, 2, 3\}$ with $p_X(k) = p_Y(k) = \frac{1}{3}$. $Z = X + Y \in \{2, 3, 4, 5, 6\}$. Out of $3 \times 3 = 9$ equally likely outcomes:

- $p_Z(2) = \mathbb{P}((1, 1)) = \frac{1}{9}$
- $p_Z(3) = \mathbb{P}(\{(1, 2), (2, 1)\}) = \frac{2}{9}$
- $p_Z(4) = \mathbb{P}(\{(1, 3), (2, 2), (3, 1)\}) = \frac{3}{9} = \frac{1}{3}$
- $p_Z(5) = \mathbb{P}(\{(2, 3), (3, 2)\}) = \frac{2}{9}$
- $p_Z(6) = \mathbb{P}(\{(3, 3)\}) = \frac{1}{9}$

3. Sum of Independent Poisson RVs Since $X \sim \text{Poisson}(2)$ and $Y \sim \text{Poisson}(3)$ are independent, $Z = X + Y \sim \text{Poisson}(2 + 3 = 5)$.

$$\mathbb{P}(Z = 1) = \frac{e^{-5} \cdot 5^1}{1!} = 5e^{-5} \approx \mathbf{0.0337}$$

4. Maximum of Two Geometric RVs For $p = 0.5$, the CDF is $F_X(k) = \mathbb{P}(X \leq k) = 1 - (1 - 0.5)^k = 1 - 0.5^k$. For $k = 2$: $F_X(2) = 1 - 0.25 = 0.75$.

$$F_W(2) = \mathbb{P}(\max(X, Y) \leq 2) = F_X(2) \cdot F_Y(2) = (0.75) \cdot (0.75) = 0.5625 = \frac{9}{16}$$

5. Data Science Application: Server Latency Redundancy Minimum For $V = \min(X, Y)$, evaluate survival function $\mathbb{P}(V > v) = \mathbb{P}(X > v)^2$:

- $\mathbb{P}(V > 0) = 1$

- $\mathbb{P}(V > 1) = \mathbb{P}(X > 1)^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$
- $\mathbb{P}(V > 2) = \mathbb{P}(X > 2)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$
- $\mathbb{P}(V > 3) = 0$

Computing individual PMF values:

- $p_V(1) = \mathbb{P}(V > 0) - \mathbb{P}(V > 1) = 1 - \frac{4}{9} = \frac{5}{9}$
- $p_V(2) = \mathbb{P}(V > 1) - \mathbb{P}(V > 2) = \frac{4}{9} - \frac{1}{9} = \frac{3}{9} = \frac{1}{3}$
- $p_V(3) = \mathbb{P}(V > 2) - \mathbb{P}(V > 3) = \frac{1}{9} - 0 = \frac{1}{9}$

Week 3: Expectation, Covariance, Correlation & Probability Bounds

Detailed Solutions

1. Linearity of Expectation For a single fair die $X_i \in \{1, \dots, 6\}$, $\mathbb{E}[X_i] = \frac{1+2+3+4+5+6}{6} = 3.5$. By Linearity of Expectation (holding regardless of independence):

$$\mathbb{E}[S] = \mathbb{E}\left[\sum_{i=1}^{10} X_i\right] = \sum_{i=1}^{10} \mathbb{E}[X_i] = 10 \times 3.5 = \mathbf{35}$$

2. Covariance Calculation from Joint PMF Marginal probabilities: $p_X(0) = 0.2 + 0.3 = 0.5$, $p_X(1) = 0.1 + 0.4 = 0.5$. $p_Y(1) = 0.2 + 0.1 = 0.3$, $p_Y(2) = 0.3 + 0.4 = 0.7$.

- $\mathbb{E}[X] = 0(0.5) + 1(0.5) = \mathbf{0.5}$
 - $\mathbb{E}[Y] = 1(0.3) + 2(0.7) = 0.3 + 1.4 = \mathbf{1.7}$
 - $\mathbb{E}[XY] = (0)(1)(0.2) + (0)(2)(0.3) + (1)(1)(0.1) + (1)(2)(0.4) = 0.1 + 0.8 = \mathbf{0.9}$
- $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.9 - (0.5)(1.7) = 0.9 - 0.85 = \mathbf{0.05}$

3. Correlation Coefficient

- **(a) Variances:**

$$\mathbb{E}[X^2] = 0^2(0.5) + 1^2(0.5) = 0.5 \implies \text{Var}(X) = 0.5 - (0.5)^2 = \mathbf{0.25}$$

$$\mathbb{E}[Y^2] = 1^2(0.3) + 2^2(0.7) = 0.3 + 2.8 = 3.1 \implies \text{Var}(Y) = 3.1 - (1.7)^2 = 3.1 - 2.89 = \mathbf{0.21}$$

- **(b) Pearson Correlation Coefficient $\rho_{X,Y}$:**

$$\sigma_X = \sqrt{0.25} = 0.5, \quad \sigma_Y = \sqrt{0.21} \approx 0.4583$$

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{0.05}{0.5 \times 0.4583} \approx \mathbf{0.2182}$$

4. Markov's Inequality Tail Bound Since visits $X \geq 0$ is a non-negative random variable with $\mathbb{E}[X] = 500$:

$$\mathbb{P}(X \geq 2000) \leq \frac{\mathbb{E}[X]}{2000} = \frac{500}{2000} = \mathbf{0.25} = \mathbf{25\%}$$

5. Data Science Application: Algorithmic Trading Risk Bound Given $\mu = 100$ and $\sigma = 20$: The interval $\$60 < X < \140 corresponds to $|X - 100| < 40 = 2\sigma$ ($k = 2$). By Chebyshev's Inequality:

$$\mathbb{P}(|X - 100| \geq 40) \leq \frac{1}{k^2} = \frac{1}{2^2} = 0.25$$

Taking the complement:

$$\mathbb{P}(60 < X < 140) = 1 - \mathbb{P}(|X - 100| \geq 40) \geq 1 - 0.25 = \mathbf{0.75} = \mathbf{75\%}$$

The probability that daily return falls between \$60 and \$140 is at least 75%.

Week 4: Continuous Random Variables, CDF, PDF & Standard Distributions

Detailed Solutions

1. PDF Normalization and Probability

- (a) Constant c :

$$\int_0^2 cx^2 dx = c \left[\frac{x^3}{3} \right]_0^2 = c \left(\frac{8}{3} \right) = 1 \implies c = \frac{3}{8}$$

- (b) Probability $\mathbb{P}(1 \leq X \leq 2)$:

$$\mathbb{P}(1 \leq X \leq 2) = \int_1^2 \frac{3}{8}x^2 dx = \frac{3}{8} \left[\frac{x^3}{3} \right]_1^2 = \frac{1}{8}(8 - 1) = \frac{7}{8} = 0.875$$

2. CDF to PDF Conversion Given $F_X(x) = 1 - e^{-3x}$ for $x \geq 0$:

- **Distribution:** Exponential($\lambda = 3$).
- **PDF $f_X(x)$:** $f_X(x) = \frac{d}{dx}(1 - e^{-3x}) = 3e^{-3x}$ for $x \geq 0$.
- **Expected Value:** $\mathbb{E}[X] = \frac{1}{\lambda} = \frac{1}{3}$.

3. Exponential Memoryless Property By the memoryless property of the exponential distribution:

$$\mathbb{P}(T > 4 \mid T > 2) = \mathbb{P}(T > 4 - 2) = \mathbb{P}(T > 2)$$

$$\mathbb{P}(T > 2) = 1 - F_T(2) = 1 - (1 - e^{-0.5 \times 2}) = e^{-1} \approx 0.3679$$

4. Standard Normal Z-Score Calculation For $X \sim \mathcal{N}(50, 100)$, $\mu = 50$, $\sigma = \sqrt{100} = 10$.

$$Z = \frac{X - 50}{10} \implies \mathbb{P}(X \geq 65) = \mathbb{P}\left(Z \geq \frac{65 - 50}{10}\right) = \mathbb{P}(Z \geq 1.5)$$

$$\mathbb{P}(Z \geq 1.5) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668 = 6.68\%$$

5. Data Science Application: Feature Normalization Range For $X \sim \text{Uniform}(10, 50)$:

- (1) Mean & Standard Deviation:

$$\mu_X = \frac{10 + 50}{2} = 30, \quad \sigma_X = \frac{50 - 10}{\sqrt{12}} = \frac{40}{2\sqrt{3}} = \frac{20}{\sqrt{3}} \approx 11.547$$

- (2) 1-Sigma Interval Probability:

$$\mathbb{P}(\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X) = \frac{(\mu_X + \sigma_X) - (\mu_X - \sigma_X)}{50 - 10} = \frac{2\sigma_X}{40} = \frac{\sigma_X}{20} = \frac{20/\sqrt{3}}{20} = \frac{1}{\sqrt{3}} \approx 57.74\%$$

Week 5: Functions of Continuous RVs, Expectations & Mixed Distributions

Detailed Solutions

1. CDF Transformation Method Given $X \sim \text{Uniform}(0, 1)$ ($F_X(x) = x$ for $x \in [0, 1]$) and $Y = -\ln(X)$ for $y > 0$:

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(-\ln(X) \leq y) = \mathbb{P}(\ln(X) \geq -y) = \mathbb{P}(X \geq e^{-y}) = 1 - F_X(e^{-y}) = 1 - e^{-y}$$

Differentiating to get PDF:

$$f_Y(y) = \frac{d}{dy}(1 - e^{-y}) = e^{-y} \quad \text{for } y > 0$$

Y follows an **Exponential distribution with rate $\lambda = 1$** ($Y \sim \text{Exponential}(1)$).

2. Linear Transformation of Normal RV For $X \sim \mathcal{N}(0, 1)$ and $Y = 3X + 5$:

- $\mathbb{E}[Y] = 3\mathbb{E}[X] + 5 = 3(0) + 5 = \mathbf{5}$.
- $\text{Var}(Y) = 3^2 \text{Var}(X) = 9(1) = \mathbf{9}$.

$Y \sim \mathcal{N}(5, 9)$ with PDF:

$$f_Y(y) = \frac{1}{3\sqrt{2\pi}} \exp\left(-\frac{(y-5)^2}{18}\right)$$

3. Continuous LOTUS Expectation For $f_X(x) = 2x$ on $[0, 1]$:

$$\mathbb{E}[X^2] = \int_0^1 x^2(2x) dx = \int_0^1 2x^3 dx = \left[\frac{2x^4}{4}\right]_0^1 = \frac{2}{4} = \mathbf{0.5}$$

4. Variance of Uniform RV For $X \sim \text{Uniform}(0, 4)$, $f_X(x) = \frac{1}{4}$ on $[0, 4]$:

- $\mathbb{E}[X] = \int_0^4 x \left(\frac{1}{4}\right) dx = \left[\frac{x^2}{8}\right]_0^4 = 2$.
- $\mathbb{E}[X^2] = \int_0^4 x^2 \left(\frac{1}{4}\right) dx = \left[\frac{x^3}{12}\right]_0^4 = \frac{64}{12} = \frac{16}{3}$.

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{16}{3} - 2^2 = \frac{16}{3} - \frac{12}{3} = \frac{4}{3} \approx \mathbf{1.3333}$$

5. Data Science Application: Gaussian Mixture Model Marginal PDF By the Law of Total Probability for mixed distributions:

$$f_X(x) = f_{X|N}(x | 0)p_N(0) + f_{X|N}(x | 1)p_N(1)$$

Substituting parameters ($\mu_0 = 50, \sigma_0 = 10$ and $\mu_1 = 300, \sigma_1 = 20$):

$$f_X(x) = \mathbf{0.95} \left(\frac{1}{10\sqrt{2\pi}} e^{-\frac{(x-50)^2}{200}} \right) + \mathbf{0.05} \left(\frac{1}{20\sqrt{2\pi}} e^{-\frac{(x-300)^2}{800}} \right)$$

Week 6: Joint Continuous RVs, Marginal Densities & Conditional Densities

Detailed Solutions

1. Joint Density Normalization Constant

$$\int_0^1 \int_0^2 cxy \, dy \, dx = c \left(\int_0^1 x \, dx \right) \left(\int_0^2 y \, dy \right) = c \left(\frac{1}{2} \right) (2) = c = 1 \implies \mathbf{c = 1}$$

$$\mathbb{P}(X \leq 0.5, Y \leq 1) = \int_0^{0.5} x \, dx \int_0^1 y \, dy = \left[\frac{x^2}{2} \right]_0^{0.5} \left[\frac{y^2}{2} \right]_0^1 = \left(\frac{1}{8} \right) \left(\frac{1}{2} \right) = \frac{1}{16} = \mathbf{0.0625}$$

2. Marginal Densities Computation Using $c = 1$:

- (1) Marginal Density $f_X(x)$:

$$f_X(x) = \int_0^2 xy \, dy = x \left[\frac{y^2}{2} \right]_0^2 = \mathbf{2x} \quad \text{for } \mathbf{0 \leq x \leq 1}$$

- (2) Marginal Density $f_Y(y)$:

$$f_Y(y) = \int_0^1 xy \, dx = y \left[\frac{x^2}{2} \right]_0^1 = \frac{y}{2} \quad \text{for } \mathbf{0 \leq y \leq 2}$$

- (3) Independence Check:

$$f_X(x) \cdot f_Y(y) = (2x) \left(\frac{y}{2} \right) = xy = f_{X,Y}(x, y)$$

Since the product of marginals equals the joint PDF everywhere, X and Y are **INDEPENDENT**.

3. Triangular Region Joint PDF For $f_{X,Y}(x, y) = 2$ on region $0 \leq y \leq x \leq 1$:

- $f_X(x) = \int_0^x 2 \, dy = \mathbf{2x} \quad \text{for } \mathbf{0 \leq x \leq 1}.$
- $f_Y(y) = \int_y^1 2 \, dx = \mathbf{2(1 - y)} \quad \text{for } \mathbf{0 \leq y \leq 1}.$

4. Conditional Density Calculation For $0 \leq y \leq x$:

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \frac{2}{2x} = \frac{1}{x} \quad \text{for } \mathbf{0 \leq y \leq x}$$

(Notice $Y | X = x \sim \text{Uniform}(0, x)$). Conditional Expectation:

$$\mathbb{E}[Y | X = x] = \int_0^x y \left(\frac{1}{x} \right) dy = \frac{1}{x} \left[\frac{y^2}{2} \right]_0^x = \frac{x}{2}$$

5. Data Science Application: Bivariate Normal Feature Regression Substitute values $\mu_X = 0, \mu_Y = 0, \sigma_X = 1, \sigma_Y = 1, \rho = 0.8$ into the linear regression function:

$$\mathbb{E}[Y | X = x] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X) = 0 + 0.8 \left(\frac{1}{1} \right) (x - 0) = \mathbf{0.8x}$$

For feature observation $X = 2.5$:

$$\mathbb{E}[Y | X = 2.5] = 0.8(2.5) = \mathbf{2.0}$$

Week 7: Statistics from Samples & The Law of Large Numbers

Detailed Solutions

1. Sample Mean Standard Error For $n = 100, \mu = 80, \sigma^2 = 400$:

- **(1) Expectation:** $\mathbb{E}[\bar{X}_{100}] = \mu = 80$.
- **(2) Standard Error:** $\text{Var}(\bar{X}_{100}) = \frac{\sigma^2}{n} = \frac{400}{100} = 4 \implies \text{SE}(\bar{X}_{100}) = \sqrt{4} = 2$.

2. Chebyshev Sample Size Determination for WLLN We want $\mathbb{P}(|\bar{X}_n - \mu| \geq 0.5) \leq 0.05$. By Chebyshev's Inequality:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}$$

Substitute $\sigma^2 = 4, \epsilon = 0.5$:

$$\frac{4}{n(0.5)^2} \leq 0.05 \implies \frac{4}{0.25n} \leq 0.05 \implies \frac{16}{n} \leq 0.05 \implies n \geq \frac{16}{0.05} = 320$$

A minimum sample size of $n = 320$ is required.

3. Empirical CDF Evaluation Sample of 5 data points sorted: $[1.8, 2.1, 4.5, 4.5, 6.0]$.

- **(1) $\hat{F}_5(4.0)$:** Observations ≤ 4.0 are $\{1.8, 2.1\}$ (2 points).

$$\hat{F}_5(4.0) = \frac{2}{5} = 0.4$$

- **(2) $\hat{F}_5(4.5)$:** Observations ≤ 4.5 are $\{1.8, 2.1, 4.5, 4.5\}$ (4 points).

$$\hat{F}_5(4.5) = \frac{4}{5} = 0.8$$

4. Sum of i.i.d. Uniform RVs For $X_i \sim \text{Uniform}(0, 1)$, $\mathbb{E}[X_i] = \frac{1}{2}$ and $\text{Var}(X_i) = \frac{1}{12}$. For $S = \sum_{i=1}^{12} X_i$:

- $\mathbb{E}[S] = 12 \times \left(\frac{1}{2}\right) = 6$.
- $\text{Var}(S) = 12 \times \left(\frac{1}{12}\right) = 1$.

5. Data Science Application: Monte Carlo Pi Estimation Bounds Each indicator $I_i \sim \text{Bernoulli}(p = \pi/4 \approx 0.7854)$ has variance $\text{Var}(I_i) = p(1-p) = 0.7854(0.2146) \approx 0.1685$. We want $\mathbb{P}(|\hat{p}_n - p| \geq 0.01) \leq 0.10$. By Chebyshev's Inequality:

$$\frac{\text{Var}(I_i)}{n(0.01)^2} \leq 0.10 \implies \frac{0.1685}{0.0001n} \leq 0.10 \implies \frac{1685}{n} \leq 0.10 \implies n \geq \frac{1685}{0.10} = 16850$$

A minimum of $n = 16,850$ Monte Carlo trials is needed.

Week 8: Moment Generating Functions & The Central Limit Theorem

Detailed Solutions

1. MGF Derivative Moment Extraction Given $M_X(t) = (1 - 2t)^{-1}$:

- **(1) Expected Value $\mathbb{E}[X]$:**

$$M'_X(t) = -1(1 - 2t)^{-2}(-2) = 2(1 - 2t)^{-2} \implies \mathbb{E}[X] = M'_X(0) = 2(1)^{-2} = 2$$

- **(2) Variance** $\text{Var}(X)$:

$$M_X''(t) = 2(-2)(1-2t)^{-3}(-2) = 8(1-2t)^{-3} \implies \mathbb{E}[X^2] = M_X''(0) = 8(1)^{-3} = 8$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 8 - 2^2 = 8 - 4 = 4$$

- 2. MGF of Independent Sums** For $X \sim \text{Poisson}(3)$ and $Y \sim \text{Poisson}(5)$:

$$M_X(t) = \exp(3(e^t - 1)), \quad M_Y(t) = \exp(5(e^t - 1))$$

By MGF factorization for independent variables:

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t) = \exp(3(e^t - 1)) \cdot \exp(5(e^t - 1)) = \exp(8(e^t - 1))$$

By the uniqueness property of MGFs, this is the MGF of **Poisson(8)**. Thus $X + Y \sim \text{Poisson}(8)$.

- 3. CLT Probability Approximation for Sample Means** For $n = 64, \mu = 40, \sigma = 8$:

$$\text{SE}(\bar{X}_{64}) = \frac{\sigma}{\sqrt{n}} = \frac{8}{\sqrt{64}} = \frac{8}{8} = 1$$

By CLT, $\bar{X}_{64} \approx \mathcal{N}(40, 1^2)$:

$$\mathbb{P}(\bar{X}_{64} > 41.5) = \mathbb{P}\left(Z > \frac{41.5 - 40}{1}\right) = \mathbb{P}(Z > 1.5) = 1 - \Phi(1.5) = 1 - 0.9332 = \mathbf{0.0668} = \mathbf{6.68\%}$$

- 4. CLT Approximation for Binomial Counts** For Binomial sum $S = \sum_{i=1}^{100} X_i$ with $n = 100, p = 0.5$:

$$\mu_S = np = 50, \quad \sigma_S = \sqrt{np(1-p)} = \sqrt{100(0.5)(0.5)} = \sqrt{25} = 5$$

Using continuity correction for $\mathbb{P}(45 \leq S \leq 55) = \mathbb{P}(44.5 \leq S \leq 55.5)$:

$$Z_1 = \frac{44.5 - 50}{5} = -1.1, \quad Z_2 = \frac{55.5 - 50}{5} = 1.1$$

$$\begin{aligned} \mathbb{P}(-1.1 \leq Z \leq 1.1) &= \Phi(1.1) - \Phi(-1.1) = \Phi(1.1) - (1 - \Phi(1.1)) = 2\Phi(1.1) - 1 \\ &= 2(0.8643) - 1 = 1.7286 - 1 = \mathbf{0.7286} = \mathbf{72.86\%} \end{aligned}$$

- 5. Data Science Application: A/B Test Conversion Sample Size** Given $\sigma = 0.30$ and target standard error $\text{SE}(\bar{X}_n) \leq 0.005$:

$$\frac{\sigma}{\sqrt{n}} \leq 0.005 \implies \frac{0.30}{\sqrt{n}} \leq 0.005 \implies \sqrt{n} \geq \frac{0.30}{0.005} = 60 \implies n \geq 60^2 = \mathbf{3600}$$

A minimum of $n = 3,600$ visitor sessions is required per variant.

Week 9: Parameter Estimation: MME, MLE & Confidence Intervals

Detailed Solutions

- 1. Unbiasedness Verification** For $\hat{\mu} = \frac{1}{4}X_1 + \frac{1}{2}X_2 + \frac{1}{4}X_3$:

- **Expectation:**

$$\mathbb{E}[\hat{\mu}] = \frac{1}{4}\mathbb{E}[X_1] + \frac{1}{2}\mathbb{E}[X_2] + \frac{1}{4}\mathbb{E}[X_3] = \frac{1}{4}\mu + \frac{1}{2}\mu + \frac{1}{4}\mu = \mu$$

Since $\mathbb{E}[\hat{\mu}] = \mu$, $\hat{\mu}$ is an **unbiased estimator**.

- **Variance:** By independence of sample observations:

$$\text{Var}(\hat{\mu}) = \left(\frac{1}{4}\right)^2 \sigma^2 + \left(\frac{1}{2}\right)^2 \sigma^2 + \left(\frac{1}{4}\right)^2 \sigma^2 = \left(\frac{1}{16} + \frac{4}{16} + \frac{1}{16}\right) \sigma^2 = \frac{3}{8} \sigma^2 = 0.375 \sigma^2$$

2. MME Parameter Estimation For $X \sim \text{Exponential}(\lambda)$, population first moment is $\mathbb{E}[X] = \frac{1}{\lambda}$. Set sample moment equal to population moment:

$$\bar{X}_n = \frac{1}{\lambda} \implies \hat{\lambda}_{\text{MME}} = \frac{1}{\bar{X}_n}$$

3. MLE Parameter Estimation For $X_i \sim \text{Poisson}(\lambda)$, joint likelihood:

$$L(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod x_i!}$$

Log-likelihood function:

$$\ell(\lambda) = \ln L(\lambda) = -n\lambda + \left(\sum_{i=1}^n x_i\right) \ln(\lambda) - \sum_{i=1}^n \ln(x_i!)$$

Differentiate with respect to λ and set to 0:

$$\frac{d\ell}{d\lambda} = -n + \frac{\sum x_i}{\lambda} = 0 \implies n\lambda = \sum_{i=1}^n x_i \implies \hat{\lambda}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{X}_n$$

4. 95% Confidence Interval Construction Given $n = 100$, $\bar{X} = 250$, $\sigma = 40$, $z_{0.025} = 1.96$:

$$\text{Margin of Error } E = z_{0.025} \frac{\sigma}{\sqrt{n}} = 1.96 \left(\frac{40}{\sqrt{100}} \right) = 1.96(4) = 7.84$$

$$95\% \text{ CI} = [250 - 7.84, 250 + 7.84] = [\$242.16, \$257.84]$$

5. Data Science Application: Logistic Regression MLE Loss For $Y_i \sim \text{Bernoulli}(p_i)$ with $p_i = \sigma(\mathbf{w}^T \mathbf{x}_i)$:

$$L(\mathbf{w}) = \prod_{i=1}^N p_i^{y_i} (1 - p_i)^{1-y_i}$$

Taking log-likelihood:

$$\ell(\mathbf{w}) = \sum_{i=1}^N [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)]$$

Multiplying by -1 yields the standard **Binary Cross-Entropy Loss** minimized by ML algorithms:

$$J(\mathbf{w}) = -\ell(\mathbf{w}) = -\sum_{i=1}^N [y_i \ln(\mathbf{p}_i) + (1 - y_i) \ln(1 - \mathbf{p}_i)]$$

Week 10: Bayesian Estimation: Prior, Posterior, MAP & Bayes Risk

Detailed Solutions

1. Beta-Binomial Conjugate Posterior Given prior $p \sim \text{Beta}(\alpha = 2, \beta = 2)$ and likelihood for $k = 4$ successes in $n = 5$ trials:

- **(1) Posterior Distribution:** By conjugate update rules, $p \mid k \sim \text{Beta}(\alpha + k, \beta + n - k) = \text{Beta}(2 + 4, 2 + 5 - 4) = \text{Beta}(6, 3)$.

- **(2) MAP Estimator $\hat{p}_{\text{MAP}}$:** Mode of $\text{Beta}(6, 3)$:

$$\hat{p}_{\text{MAP}} = \frac{\alpha' - 1}{\alpha' + \beta' - 2} = \frac{6 - 1}{6 + 3 - 2} = \frac{5}{7} \approx 0.7143$$

- **(3) Bayes Posterior Mean Estimator $\hat{p}_{\text{Bayes}}$:**

$$\hat{p}_{\text{Bayes}} = \mathbb{E}[p \mid k] = \frac{\alpha'}{\alpha' + \beta'} = \frac{6}{6 + 3} = \frac{6}{9} = \frac{2}{3} \approx 0.6667$$

2. Normal-Normal Conjugate Model For $X \mid \mu \sim \mathcal{N}(\mu, \sigma^2 = 4)$ and prior $\mu \sim \mathcal{N}(\mu_0 = 0, \sigma_0^2 = 1)$:

- Posterior Precision: $\frac{1}{\sigma_{\text{post}}^2} = \frac{1}{\sigma_0^2} + \frac{1}{\sigma^2} = \frac{1}{1} + \frac{1}{4} = \frac{5}{4} \implies \sigma_{\text{post}}^2 = 0.8$.
- Posterior Mean:

$$\mathbb{E}[\mu \mid X = 5] = \sigma_{\text{post}}^2 \left(\frac{\mu_0}{\sigma_0^2} + \frac{x}{\sigma^2} \right) = 0.8 \left(0 + \frac{5}{4} \right) = 0.8(1.25) = 1.0$$

3. Uniform Prior MAP vs MLE Comparison For $X \sim \text{Binomial}(10, p)$ with uniform prior $p \sim \text{Beta}(1, 1)$: Posterior: $p \mid x \sim \text{Beta}(1 + x, 1 + 10 - x) = \text{Beta}(x + 1, 11 - x)$.

$$\hat{p}_{\text{MAP}} = \frac{(x + 1) - 1}{(x + 1) + (11 - x) - 2} = \frac{x}{10} = \hat{\mathbf{p}}_{\text{MLE}}$$

4. Laplace Prior and L_1 Regularization Equivalence For prior $f(w_j) = \frac{\lambda}{2} e^{-\lambda|w_j|}$, log-prior is $\ln f(\mathbf{w}) = \sum_j \ln \left(\frac{\lambda}{2} \right) - \lambda \sum_j |w_j| = C - \lambda \|\mathbf{w}\|_1$. Log-posterior objective:

$$\ln f(\mathbf{w} \mid \mathbf{x}, y) = \ln f(y \mid \mathbf{x}, \mathbf{w}) + \ln f(\mathbf{w}) = \ln L(\mathbf{w}) - \lambda \|\mathbf{w}\|_1 + C$$

Maximizing the log-posterior is mathematically identical to minimizing the negative log-likelihood plus an L_1 **Lasso Penalty term** $\lambda \|\mathbf{w}\|_1$.

5. Data Science Application: Bayesian AB Testing Click Probability Prior $p \sim \text{Beta}(1, 1)$, observations $n = 100, k = 15$:

- **Posterior:** $\text{Beta}(1 + 15, 1 + 85) = \text{Beta}(16, 86)$.
- **Posterior Mean Estimate:**

$$\mathbb{E}[p \mid \text{data}] = \frac{16}{16 + 86} = \frac{16}{102} = \frac{8}{51} \approx 0.1569 = 15.69\%$$

Week 11: Hypothesis Testing: Size, Power, Z-Test & p-Values

Detailed Solutions

1. Type I and Type II Error Identification

- **(1) Type I Error (α):** False Positive Concluding the patient has the disease (H_1) when they are actually healthy (H_0).
- **(2) Type II Error (β):** False Negative Concluding the patient is healthy (H_0) when they actually have the disease (H_1). A **Type II Error is much more dangerous** because a sick patient receives no medical treatment!

2. One-Sample Right-Tailed Z-Test

- **(1) Hypotheses:** $H_0 : \mu \leq 48$ vs. $H_1 : \mu > 48$.
- **(2) Test Statistic:**

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{64}} = \frac{12}{8} = 1.5 \implies Z_{\text{calc}} = \frac{51 - 48}{1.5} = \frac{3}{1.5} = 2.0$$

- **(3) Decision:** At $\alpha = 0.05$, critical value is $z_{0.05} = 1.645$. Since $Z_{\text{calc}} = 2.0 > 1.645$, we **REJECT** H_0 . There is strong evidence that delivery time exceeds 48 hours.

3. p-Value Calculation For right-tailed $Z_{\text{calc}} = 2.0$:

$$p\text{-value} = \mathbb{P}(Z \geq 2.0) = 1 - \Phi(2.0) = 1 - 0.9772 = 0.0228 = 2.28\%$$

4. Two-Tailed Z-Test with p-Value $H_0 : \mu = 500$ vs $H_1 : \mu \neq 500$.

- Test statistic: $Z_{\text{calc}} = \frac{496-500}{20/\sqrt{100}} = \frac{-4}{2} = -2.0$.
- Two-tailed p-value: $2 \times \mathbb{P}(Z \geq |-2.0|) = 2 \times (1 - \Phi(2.0)) = 2 \times 0.0228 = 0.0456 = 4.56\%$.
- Decision at $\alpha = 0.05$: Since $p\text{-value} = 0.0456 \leq 0.05$, we **REJECT** H_0 . The boxes are mis-filled.

5. Data Science Application: A/B Test Significance Decision For two-tailed $Z_{\text{calc}} = 2.33$:

$$p\text{-value} = 2 \times (1 - \Phi(2.33)) = 2 \times (1 - 0.9901) = 2 \times 0.0099 = 0.0198 = 1.98\%$$

At significance level $\alpha = 0.01$ (1%): Since $p\text{-value} = 0.0198 > 0.01$, we **FAIL TO REJECT** H_0 at the 1% level (the uplift is not statistically significant at $\alpha = 0.01$, though it would be at $\alpha = 0.05$).

Week 12: Advanced Hypothesis Testing, LRT & Goodness-of-Fit

Detailed Solutions

1. One-Sample t-Test Calculation For $n = 16$, $\bar{X} = 104$, $S = 8$:

- **(1) Degrees of Freedom:** $df = n - 1 = 16 - 1 = 15$.
- **(2) Test Statistic:** $SE = \frac{S}{\sqrt{n}} = \frac{8}{\sqrt{16}} = 2 \implies T_{\text{calc}} = \frac{104-100}{2} = \frac{4}{2} = 2.0$.
- **(3) Decision at $\alpha = 0.05$:** Two-tailed critical value $t_{0.025,15} = 2.131$. Since $|T_{\text{calc}}| = 2.0 < 2.131$, we **FAIL TO REJECT** H_0 . The sample mean does not differ significantly from 100.

2. Two-Sample Pooled t-Test Given $n_1 = 10$, $\bar{X}_1 = 15$, $S_1^2 = 4$ and $n_2 = 10$, $\bar{X}_2 = 12$, $S_2^2 = 6$:

- Pooled Variance:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} = \frac{9(4) + 9(6)}{10 + 10 - 2} = \frac{36 + 54}{18} = \frac{90}{18} = 5$$

- Standard Error: $SE = \sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{5 \left(\frac{1}{10} + \frac{1}{10} \right)} = \sqrt{5 \times 0.2} = \sqrt{1} = 1$.
- Test Statistic: $T_{\text{calc}} = \frac{15-12}{1} = 3.0$.

3. Chi-Squared Goodness of Fit (Die Fairness Test) $N = 60, E_i = 10$ for all $i \in \{1, \dots, 6\}$. Observed $O = [15, 7, 8, 12, 10, 8]$:

$$\chi_{\text{calc}}^2 = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i} = \frac{(15 - 10)^2 + (7 - 10)^2 + (8 - 10)^2 + (12 - 10)^2 + (10 - 10)^2 + (8 - 10)^2}{10}$$

$$\chi_{\text{calc}}^2 = \frac{25 + 9 + 4 + 4 + 0 + 4}{10} = \frac{46}{10} = \mathbf{4.6}$$

At $\alpha = 0.05$ with $df = 6 - 1 = 5$, critical value $\chi_{0.05,5}^2 = 11.07$. Since $\chi_{\text{calc}}^2 = 4.6 < 11.07$, we **FAIL TO REJECT** H_0 . The die is consistent with being fair.

4. Wilks' Likelihood Ratio Test Stat Given likelihood ratio $\Lambda = 0.05$:

$$W = -2 \ln \Lambda = -2 \ln(0.05) = -2(-2.9957) \approx \mathbf{5.9914}$$

5. Comprehensive Course Review: χ^2 Independence Test Compute cell contributions $\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$:

- Free / Retained: $\frac{(80-90)^2}{90} = \frac{100}{90} \approx 1.111$
- Free / Churned: $\frac{(40-30)^2}{30} = \frac{100}{30} \approx 3.333$
- Premium / Retained: $\frac{(70-60)^2}{60} = \frac{100}{60} \approx 1.667$
- Premium / Churned: $\frac{(10-20)^2}{20} = \frac{100}{20} = 5.000$

$$\chi_{\text{calc}}^2 = 1.111 + 3.333 + 1.667 + 5.000 = \mathbf{11.111}$$

With $df = (2 - 1)(2 - 1) = 1$, critical value $\chi_{0.05,1}^2 = 3.841$. Since $\chi_{\text{calc}}^2 = 11.111 > 3.841$, we **REJECT** H_0 . Subscription Tier and Churn status are statistically significantly associated!